

Team Makes You Better: Evidence from Online Medical Consultation Platforms

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Abstract

Virtual teamwork has become a prevalent mode of collaboration. However, scant empirical attention has been given to the impact of virtual teamwork on individual members' performance beyond the team setting. In this paper, we fill this gap by studying virtual teams in the context of online medical consultation (OMC) platforms. Specifically, we collect data from two OMC platforms during a time period when both platforms offered individual consultations (i.e., one patient connected with one physician) but only one offered team consultations (i.e., one patient connected with a team of physicians). Using a difference-in-difference-in-differences approach, we find that team participation has a robust, positive effect on physicians' performance in individual consultations. We further explore the potential mechanisms through which team participation improves physicians' performance. The empirical results indicate that the observed performance improvement is likely driven by peer effects in team settings. In particular, we find that physicians exert more effort in individual consultations after joining a team, and that performance improvement is stronger when (1) a physician can easily observe the performance of his teammates (peers), (2) a physician has teammates (peers) who performs well in individual consultations, (3) a physician has teammates (peers) from the same hospital or the same city, or (4) a physician has a lower performance prior to joining in a team. Our findings shed new light on the boundary-spanning roles of virtual teams and provide timely insights into the ongoing development of OMC platforms.

Keywords: Virtual teams, online medical consultation, difference-in-difference-in-differences, peer effects, digital platforms

1. Introduction

Team-based work structures have long been used by organizations to facilitate knowledge sharing, improve productivity, and stimulate innovation (Hahn et al. 2008, Mortensen 2014, Agha et al. 2022). The proliferation of information and communication technologies (e.g., high-speed internet, mobile, and cloud-computing services) has enabled individuals who are geographically dispersed to work together (Malhotra et al. 2007, Wakefield et al. 2008). Compared to traditional teams, virtual teams offer greater flexibility and efficiency (Majchrzak et al. 2000, Wilson et al. 2013), which are becoming increasingly important in today's fast-evolving world.

Over the past decades, much progress has been made on how social, behavioral, and cognitive factors affect virtual team formation (Von Krogh et al. 2003, Hahn et al. 2008), coordination (e.g., Kanawattanachai and Yoo 2007, Choi et al. 2010, Kudaravalli et al. 2017), and performance (Jarvenpaa and Leidner 1999, Kayworth and Leidner 2002, Kanawattanachai and Yoo 2007, Bartelt and Dennis 2014). However, there is limited research investigating the effects of being on a virtual team on members' performance on individual tasks. It is worth noting that in practice, members of virtual teams, especially those formed based on mutual interests or shared goals (e.g., programmers contributing to open-source software development, experts in working groups or task forces), are often performing similar tasks as individuals in parallel to their team tasks. The dual commitment creates a channel for knowledge spillover (Raghuram et al. 2010) and peer effects (Horton and Zeckhauser 2018), but at the same time, it also presents nuanced tradeoffs between engagement in virtual teamwork and individual tasks. As such, it is unclear whether and under what circumstances participation in virtual teams can benefit individual members, especially in their individual tasks.

In this paper, we seek to address this question by studying the performance implications of virtual teamwork on physicians in online medical consultation (OMC) platforms. Unlike traditional face-to-face clinic visits, OMC platforms connect patients with physicians remotely via text, picture, audio, and video. By eliminating geographic and time barriers, these platforms greatly improve accessibility to high-quality

care services (Cheng et al. 2022), and hence are deemed instrumental to achieving equity in care in countries with uneven/unbalanced distribution of healthcare resources (Webster 2020, Zhou et al. 2024). Despite the consensus on their promises, how to operationalize these platforms for the maximum benefits remains an open question. A top concern pertains to the limited care resources. As more patients have adopted OMC, there are mounting complaints about long waiting times and insufficient responses from physicians (Qu 2023). In light of this, some OMC platforms have launched a team consultation feature that enables physicians to form virtual teams to jointly deliver care. The rationale is that such virtual teamwork can reduce patients' waiting time by allowing any available physician in a team to respond to patients' inquiries (Qu 2023). Moreover, the collaborative care delivery mode can facilitate knowledge sharing among team members and reduce medical errors (Weller et al. 2014). However, this perspective may be missing important considerations. In particular, since most physicians who provide OMC services already have a tight work schedule offline, it is unclear whether and how participation in team consultation would affect their performance in individual consultations. On the one hand, physicians in a consultation team would inevitably incur coordination costs (DeShon et al. 2004), which may adversely affect their responsiveness and capacity in individual consultations (Du et al. 2024). On the other hand, the presence of team members may fuel the aspiration for performance improvement, which leads to positive spillover in individual consultations.

Motivated by these practical considerations, we empirically examine the effect of team consultation on physicians' performance in individual consultations. A major threat to identification lies in the potential self-selection related to physicians' voluntary team formation: There are unobservable factors that can simultaneously affect team formation and physicians' performance in individual consultations. We address this empirical challenge by leveraging a quasi-natural experimental setting. In particular, we collect patient-physician encounter data from two leading OMC platforms in China: One implemented the team consultation feature during our study period, and the other did not. Using a difference-in-difference-in-differences (DDD) approach (Chevalier and Mayzlin 2006, Gao et al. 2024), we find that team participation has a significant positive effect on physicians' performance (measured by patient satisfaction) in individual

consultations.

Several potential mechanisms may drive the positive spillover effect. Drawing upon the characteristics of physicians' interaction on OMC platforms, we focus on two mechanisms: knowledge-sharing effects and peer effects. Our analysis suggests that the observed performance improvement associated with team participation is likely driven by peer effects that manifest through social comparison (Smith 2000) in team settings. In particular, we find that physicians exert more effort after participating in team consultation, and that performance improvement is stronger when (1) a physician can easily observe the performance of his teammates (peers), (2) a physician has teammates (peers) who performs well in individual consultations, (3) a physician has teammates (peers) from the same hospital or the same city, or (4) a physician has a lower performance prior to joining in a team. These findings consistently support the social comparison explanation.

Our study makes several contributions. First, we add to the literature on virtual teams by empirically investigating the impact of virtual teamwork on individual performance. As virtual collaboration becomes the new norm, understanding how interactions and communications within a virtual team transcend the team setting and affect individual members' performance is beneficial for both internal (individual team members) and external stakeholders (e.g., customers served by individual team members). Second, our paper extends the existing research on peer effects in offline settings. In particular, our empirical analysis provides compelling evidence that the increased transparency within a virtual team setting can motivate members to increase effort in individual tasks, which can further translate into performance improvement. Finally, our paper contributes to the design and operation of OMC platforms. Our findings shed new light on the implications of the coexistence of individual and team consultation and provide guidance on how to effectively manage the two consultation modes for maximum benefits.

The rest of this paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the research context. Section 4 introduces our identification strategy and presents the empirical findings. Section 5 delves into the potential mechanisms. Section 6 discusses the theoretical and practical insights as well as the directions for future research.

2. Literature Review

Virtual teams have been increasingly employed by organizations for strategic analysis (Maznevski and Chudoba 2000), customer service (Kirkman et al. 2004), and product development (Majchrzak et al. 2000). The existing studies largely focus on key drivers of the success and failure of virtual teams, such as trust (Jarvenpaa and Leidner 1999), leadership (Kayworth and Leidner 2002), culture (Massey et al. 2001), and team members' ability to share and apply knowledge (Faraj and Sproull 2000, Kanawattanachai and Yoo 2007, Choi et al. 2010). Our research complements this stream of literature¹ by studying the spillover effects of collaboration in virtual teams on team members' performance in individual tasks. Few papers have examined the impact of virtual team participation on individual performance. A notable exception is Mell et al. (2021), where the authors introduced the construct of temporal brokerage, a position within a team's temporal structure that bridges subgroups with little or no temporal overlap, and showed that occupying such a position in a virtual team and engaging in more coordination work than other team members decreases the quantity but increases the quality of an individual's total productive output. Our paper shares the same spirit of Mell et al. (2021) by studying the implications of teamwork that go beyond the boundaries of a focal team; however, unlike Mell et al. (2021), we focus on cognitive and behavioral forces, specifically, peer effects and knowledge sharing, rather than temporal dispersion in virtual teams. Below, we first discuss two streams of literature that are particularly relevant to the theoretical development of our work.

Peer Effects. Peer effects refer to the phenomenon whereby one individual's attitude and behavior are influenced by interactions with other members within a reference group (Manski 1993). Early research on peer effects primarily focused on offline settings. For example, Sacerdote (2001) and Zimmerman (2003) studied peer effects on academic performance. Christakis and Fowler (2007) and Trogon et al. (2008) explored peer effects on health-related behaviors. Mas and Moretti (2009) and Bellemare et al. (2010) examined peer effects in workplace productivity. Recently, the increasing penetration of the Internet and mobile services has fueled the interest in exploring peer effects in online settings. Specifically, Chen et al.

¹ See Gilson et al. (2015) for a more comprehensive literature review.

(2017) found that peer recognition can improve voluntary contributions on knowledge-sharing platforms. Bapna and Umyarov (2015) examined how peer effects influence the adoption of a paid service in an online social network. Huang et al. (2021b) found that communicating peers' assignment completion rates alleviates procrastination on massive online open courses (MOOC) platforms. However, these papers focus on peer effects among a general population in the absence of teamwork, whereas we study a setting where professionals (i.e., physicians) work both as a team and independently.

It is worth noting that previous research on peer effects within team settings (e.g., Mas and Moretti 2009, Bellemare et al. 2010) suggests that while the presence of high performers in a team may motivate teammates to invest more effort in team tasks, the resulting peer pressure may backfire. Our paper extends this stream of literature in two important ways. First, by examining the impact of team participation on physicians' performance on individual consultations (i.e., individual tasks rather than team tasks), our study sheds new light on the boundary-spanning role (Levina and Vaast 2005) of peer effects, that is, how peer effects that emerge from interactions with teammates transcend beyond the team environment and influence physician's individual consultation practices. Second, physicians in virtual teams can easily observe and track the performance of their peers (i.e., teammates) on both team tasks (team consultations) and individual tasks (individual consultations). Such a dual-channel setting can affect physicians' social comparison motives (Suls et al. 2002), their comparison targets (Beylat et al. 2024), and hence the outcome (Kruglanski et al. 2000).

Knowledge Sharing. Knowledge sharing—the provision and coordination of expertise to help and collaborate with others (Cummings 2004, Wasko and Faraj 2005)—has long been studied by scholars across different disciplines. In this literature, one strand of papers focuses on how organizational, individual, and technical factors influence individual-level knowledge sharing. For example, Cummings (2004) demonstrated that structural diversity (e.g., geographic dispersion and functional differences of team members) can help increase knowledge sharing. Robert et al. (2018) found that text communication improves (hurts) knowledge sharing in racially (gender) diverse teams. Kanawattanachai and Yoo (2007) and Choi et al. (2010) demonstrated how transactive memory systems facilitate knowledge sharing in virtual

teams.

Another strand of papers examines the impact of knowledge sharing on product development (Ozer and Vogel 2015), creativity (Majchrzak and Malhotra 2016), as well as the development of trust in virtual teams (Pangil and Chan 2014). It is also worth noting that knowledge sharing does not necessarily translate into performance improvement, especially when it involves costs and barriers (Haas and Hansen 2005, 2007). Our paper enriches the literature by exploring the implications of knowledge sharing beyond the team setting. In particular, most of the existing studies typically address team or individual behaviors separately. We, on the other hand, focus on individuals (i.e., physicians) who work both independently and cooperatively (as a team) simultaneously. Such a dual-channel setting allows us to explore more nuanced insights into the performance implications of knowledge sharing.

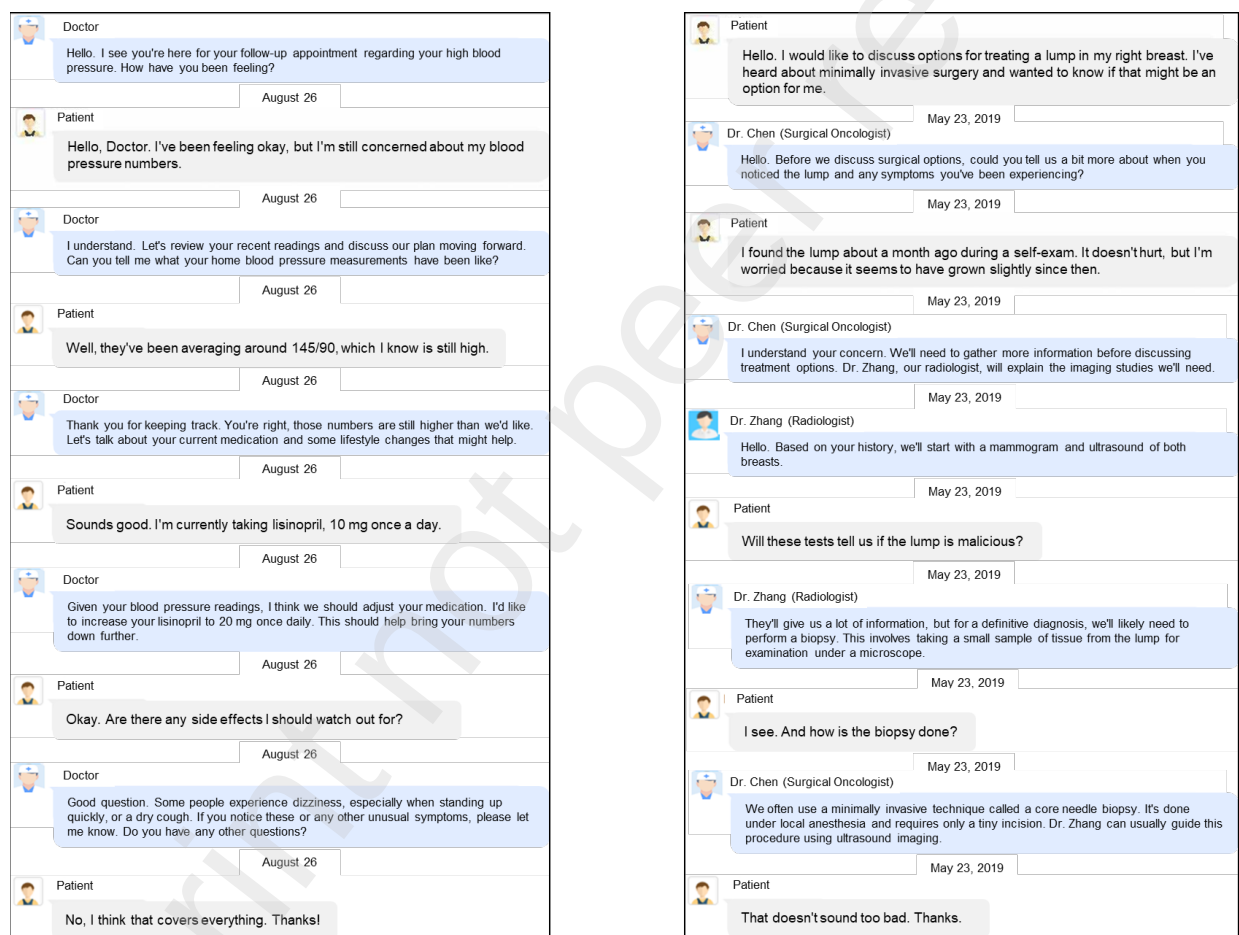
Finally, by studying virtual teamwork in OMC platforms, our paper also contributes to the nascent literature on the digital transformation of healthcare. A set of papers have studied the implications of the adoption and use of OMC. For example, Liu et al. (2020) showed that both patients' and physicians' participation in OMC platforms significantly improves patient well-being by facilitating patient-physician collaborative care and self-management support. Zhou et al. (2024) found that telehealth adoption may aggravate urban-rural health disparities. Qiao et al. (2021) found that being part of a team has a positive effect on individual physicians' reputation and demand. Our study differs from these papers in two ways. First, unlike Li et al. (2020) and Qiao et al. (2021), our empirical strategy accounts for potential self-selection in physicians' team participation. Second, we go beyond estimating the impact of virtual teamwork on physicians' performance in individual consultations by further exploring the underlying mechanisms.

3. Research Context

To understand how participation and engagement in virtual teams affect members' performance in individual tasks, we choose to focus on OMC platforms, where physicians remotely address patients' medical needs either independently (individual consultation) or work as a team (team consultation).

3.1. Overview of OMC Platforms

OMC is a form of healthcare service delivered by qualified professionals on third-party digital platforms.² In practice, once patients create a user account on an OMC platform (either via mobile apps or Internet portals), they can choose to send direct inquiries (along with medical history) and wait for responses, or book appointments for a consultation session (via text, audio, or video). Figure 1 illustrates the two modes through which patients seek advice from physicians on an OMC platform: individual consultation and team consultation (Section 3.2 provides more details about the differences between the two consultation modes).



(a) Individual Consultation

(b) Team Consultation

Figure 1. Illustration of Patient-Physician Interaction on an OMC Platform

OMC has seen increasing adoption for diagnosing and treating common diseases (e.g., cold, flu,

² While OMC falls into the umbrella term known as telehealth, a major difference is that the latter can also refer to virtual training and collaboration of care providers (Blandford et al. 2020).

and mental disorders) as well as assistance and control of chronic illnesses such as hypertension, diabetes, and cancer (Pearl and Wayling 2022). For countries with limited access and uneven distribution of healthcare resources, OMC is deemed an important complementary to the traditional care delivery channel (Webster 2020). Take China for example. During the peak of the COVID-19 pandemic, the country's leading OMC platforms hosted over 800,000 daily medical inquiries (Shepherd 2020). In the United States, the adoption rate of OMC remains much higher than pre-pandemic levels as of January 2023 (Eastburn et al. 2023).

3.2. Empirical Setup

Our empirical analysis leverages data from two leading OMC platforms in China: WeDoctor (guahao.com) and Haodf (haodf.com). Both platforms allow physicians to provide one-to-one individual consultations via text/image messaging, as well as voice and video calls (Fan et al. 2023). Table 1 provides an overview of the two platforms. Despite the difference in founding time, the two platforms are comparable in terms of active users (i.e., registered physicians and patients), consultation types, and pricing. Many physicians offer consultation services on both platforms (Fan et al. 2023). Patients can use either the app or the web interface to search for physicians by specifying diseases, locations, and/or hospitals, which would result in a list of physicians that fit the search criteria. Patients can then browse the physicians' profiles on the list and choose physicians and the consultation format (e.g., text/image messaging and voice/video call).

Table 1. Basic Information about WeDoctor and Haodf

	WeDoctor	Haodf
Headquarters	Hangzhou	Beijing
Founding time	2010	2006
Ownership	Private	Private
Number of registered physicians	270,000	260,000
Number of active patients per month	2.02 million	1.90 million
Consultation types	Image/text, voice, and video	Image/text, voice, and video
Price per image/text-based session (CNY)	50.74	55.81

Figure 2 provides an illustration of a physician's homepage (left panel) and a review page (right panel) on WeDoctor. Three types of information are provided on a physician's homepage: (1) the

physician's basic information, which includes a profile photo, name, affiliation, specialty, and a short biography; (2) experience on the platform, including the number of individual patients served and the consultation format (e.g., text/image messaging and voice/video call); and (3) reviews by patients who received individual consultations from the physician.

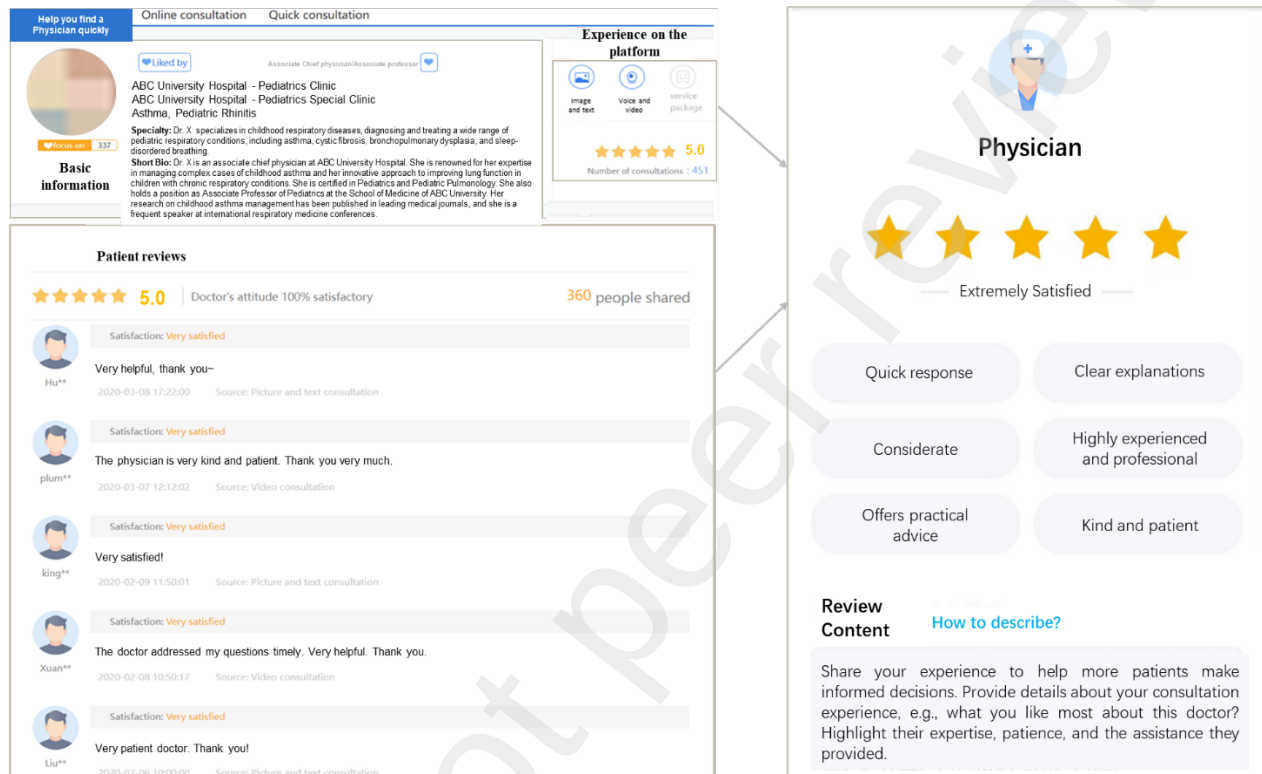


Figure 2. Illustration of a Physician's Homepage on WeDoctor

On the review page (right panel of Figure 2), a patient can give a star rating to express how satisfied she was with the physician's service (one to five stars correspond to highly unsatisfied to highly satisfied) and write detailed comments about her experience. The user interface of physicians' homepages and review pages on Haodf is similar to that of WeDoctor.

Launch of Team Consultation. As the demand for OMC continues to increase, a major challenge faced by leading OMC platforms such as WeDoctor and Haodf is how to match patients and physicians and facilitate effective and efficient care delivery. In particular, since patients' choices are largely dependent on physicians' credentials and review information (Saiffee et al. 2019), the workload among physicians on

OMC platforms is highly skewed,³ which results in the following dilemma: On the one hand, patients often complain that they cannot receive timely responses from their preferred (top-rated domain expert) physicians due to congestion; on the other hand, there is a shortage of cases for junior physicians, who need to improve their professional skills and knowledge through practice (Qu 2023).

To tackle this challenge, in May 2014, WeDoctor launched a team consultation feature that allows physicians to form teams and jointly deliver care. In practice, a well-known physician and domain expert would take the initiative to form a team and act as the team leader. Other physicians in related domains can request to join the team at any time. Compared to the individual consultation mode, team consultation can significantly reduce patients' waiting time. Specifically, once a patient reaches out to the team, any team member can directly respond to the patient's inquiry and/or request assistance from teammates. Figure 3 presents an illustrative example of a consultation team's homepage on WeDoctor. In addition to a general description of the team (e.g., the specialty), it also provides basic information about each team member. WeDoctor did not include any review system for team consultations.

Compared with individual consultation that connects one patient to one physician at a time, team consultation allows physicians (from either the same or different institutions/regions) to collaboratively provide care to a patient in need as a team, which can lead to three potential advantages. First, since any member of a team can respond to a patient request, it can reduce patients' waiting time (especially in the case of common diseases that each team member can contribute), which has been a pain point for OMC platforms (Qu 2023). Second, the combined knowledge and experience possessed by a team of physicians may better address the needs of patients with complex diseases (Liu et al. 2023), especially those with chronic diseases (Van der Eijk et al. 2013). Third, a consultation team may consist of physicians with different domain knowledge and experience, and the collaboration within the team may facilitate the dissemination and sharing of knowledge and experience (Wu 2015). Due to these potential benefits, Haodf also launched a similar team consultation feature in June 2017.

³ See https://www.sohu.com/a/364805713_115035, accessed on January 10, 2025.



微医 · 数字医院

we doctor

Q Please enter the name of the disease, department, or doctor

Search

Help you find a Physician quickly

Online consultation Quick consultation



XYZ Expert Team

Specialty:

Asthma, respiratory tract infections, chronic cough, allergic rhinitis, and cold

Team Overview:

The XYZ Expert Team is headed by Chief Physician A from the L University Hospital. It brings together four other experts, including B and C from the same department, and D and E from the M Friendship Hospital. Combining the strengths of traditional Chinese medicine and modern medicine, the team effectively treats a wide range of respiratory diseases.

Description of the team

Team style

Description of the team leader



Chief Physician

L University Hospital, Pediatrics Clinic

Chief Physician, certified in Pediatrics and Pediatric Pulmonology. Fellow of the Academy of Pediatrics. Director of the Pediatric Asthma Center at L University Hospital.

Service Activated: Not yet

Contact Dr. A



Associate Chief Physician

L University Hospital, Pediatrics Clinic

Expertise: Highly experienced in diagnosing and treating chronic respiratory conditions.

Opened service: Online consultation

Contact Dr. B



Associate Chief Physician

L University Hospital, Pediatrics Clinic

Expertise: Specialized in complex cases of childhood asthma, cystic fibrosis, and sleep-disordered breathing.

Opened service: Online consultation

Contact Dr. C



Associate Chief Physician

M Friendship Hospital, Pediatrics Clinic

Expertise: Specialized in diagnosing and treating a wide range of pediatric respiratory conditions.

Service Activated: Not yet

Contact Dr. D



Attending Physician

M Friendship Hospital, Emergency Department

Expertise: General pediatrics, blood diseases, and pediatric endocrine disorders.

Service Activated: Not yet

Contact Dr. E

Description of team members

An introduction of team consultation

What is an Expert Team?



An expert team consists of leading experts from different hospitals across the nation. These experts work together to deliver specialized care for specific diseases.

> Nationwide partnerships

> Personalized diagnostic and treatment processes

> Premium, team-based healthcare services

Articles written by the physicians in the team

Articles

1 What conditions need to be ...

Allergic diseases in children are common in childhood, and the incidence is increasing year by year.

2 209 | In addition to taking bl...

3 Look at what people are doi...

Figure 3. A Sample Homepage of a Consultation Team on WeDoctor

Despite the aforementioned advantages, however, participation in team consultation also involves coordination costs (Faraj and Sproull 2000). In particular, given physicians' busy offline schedules, it is unclear how the adoption of team consultation would affect their allocation of time and effort, and accordingly, their performance in individual consultations.

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This preprint research paper has not been peer reviewed. Electronic copy available at: <https://ssrn.com/abstract=5101552>

Quasi-Experimental Setting. The difference in the launch time of team consultation between WeDoctor and Haodf constitutes a quasi-natural experimental setting, which allows us to identify and estimate the impact of team consultation on physicians' performance. Specifically, we used the name, hospital, and department to identify physicians who registered on both platforms during the period of May 2014 to June 2017, when only WeDoctor implemented the team consultation feature. This results in a sample of 2,670 physicians. Among these 2,670 physicians, 292 joined consultation teams on WeDoctor and had been offering individual consultation before joining a consultation team; the remaining 2,378 did not adopt team consultation. Figure 4 provides an overview of the quasi-experimental setup for our analysis.

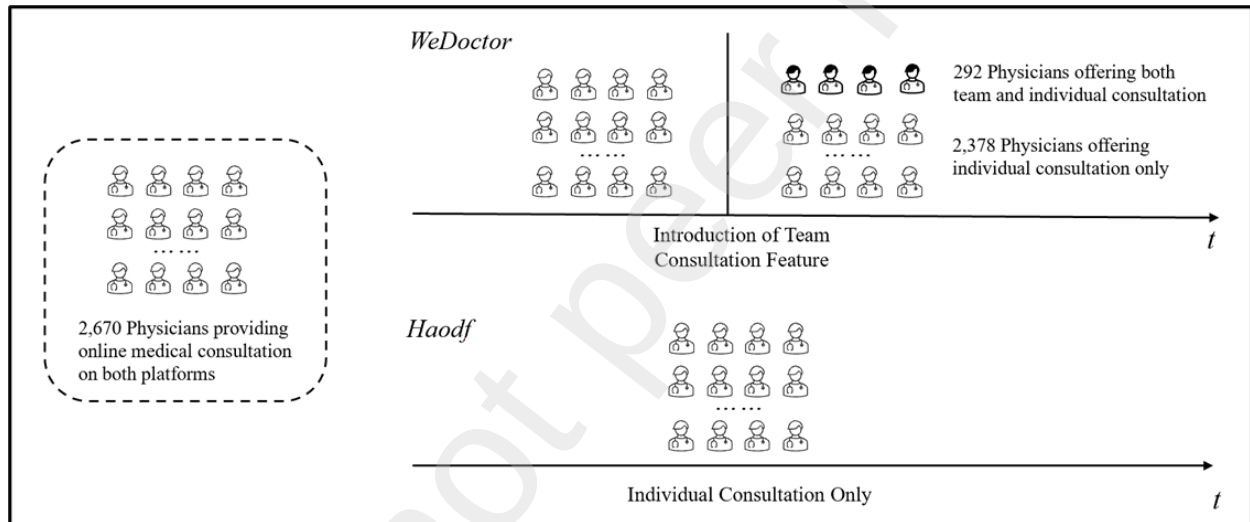


Figure 4. The Quasi-Experimental Setup Associated with the Launch of Team Consultation

We collected detailed information for each of the 2,670 physicians on both platforms, respectively. These data encompass (1) personal information, which includes name, registration date on the platform, department, specialty, professional title, recognition (domain expert or not), hospital affiliation, and hospital ranking (i.e., whether is an elite hospital or not); (2) information related to individual consultations (i.e., patient requests, physician's responses, and patient reviews); and (3) information related to team consultation (i.e., patient requests and physicians' responses). With the collected information, we can construct a panel dataset at the physician-platform-day level.

Drawing upon prior literature (Li and Hitt 2008, Burtch et al. 2018), we use the number of satisfied

reviews to measure the performance of physicians in providing individual consultations. While such a performance measure may not adequately capture the clinical outcomes of patients receiving consultations (Song et al. 2018, Saiffee et al. 2020), it is well-suited to examining the impact of team participation on individual consultations. The rationale is two-fold: first, when the number of patients a physician served through individual consultations in a given period of time (e.g., a week or a month) remains stable, i.e., the physician's productivity in providing individual consultations remains unchanged, although only a portion of the served patients would leave feedback (Li and Hitt 2008, Burtch et al. 2018), an increase in $SatiNum_{it}$ indicates better (perceived) service quality or consultation experience in the patient-physician encounter; second, when the physician's service quality remains unchanged, an increase of $SatiNum_{it}$ suggests improvement in productivity (Chintagunta et al. 2010, Saiffee et al. 2020). In either scenario, $SatiNum_{it}$ can capture the productivity and/or service quality of physician i at time t , as both measurements stand for physicians' performance in individual consultations.

4. Estimating the Effect of Team Participation

4.1. Identification Strategy

Given the quasi-natural experimental setting introduced in Section 3.2, intuitively, we could use a difference-in-differences (DD) approach to examine the impact of team participation on physicians' performance in individual consultations by comparing the performance change of physicians who joined a team (treatment group) versus that of physicians who did not join a team (control group) on WeDoctor. This leads to the following DD model:

$$Y_{it}^W = \beta_0 + \beta_1 After_{it}^W + \eta Z_{it}^W + \theta_t^W + \sigma_i^W + \varepsilon_{it}^W, \quad (1)$$

where Y_{it}^W denotes the number of satisfied reviews⁴ physician i received on day t on WeDoctor (i.e., $SatiNum_{it}^W$). $After_{it}^W$ is a binary variable, which indicates whether physician i was part of a consultation team on WeDoctor on day t . Z_{it}^W includes control variables that vary among physicians over time. θ_t^W and

⁴ In line with practice, we define reviews with four- and five-star ratings as satisfied reviews.

σ_i^W correspond to the time-fixed effects⁵ and the physician-fixed effects, respectively. ε_{it}^W is the error term that follows a standard normal distribution.

A potential concern about the DD model in Equation (1) is that it does not control for unobservable physician-specific time-varying factors that may affect physicians' performance in individual consultations as well as their decisions to join a team. As a result, the estimate of β_1 may be biased. In particular, the observed change of physicians' performance in individual consultations may coincidentally happen with the change in their domain knowledge and work experience. Also, the decision to join a team and the performance in individual consultations may both be affected by physicians' personal aspirations, such as their willingness to learn new knowledge. One way to alleviate this concern is to adopt a difference-in-difference-in-differences (DDD) estimation approach (Chevalier and Mayzlin 2006, Olden and Møen 2022, Gao et al. 2024) by taking into account physicians' performance in individual consultations on Haodf where team consultation was not implemented during our study period. Alternatively, we can run separate DD estimations on WeDoctor and Haodf. If physicians' performance in individual consultations only improves on WeDoctor but not on Haodf, it means the unobservable physician-specific time-varying factors do not constitute a threat to the identification. For expositional purposes, we adopt the DDD approach in the main analysis and include the separate DD estimations as a robustness check (see Appendix B for a detailed discussion on the estimation strategy and results).

Difference-in-Difference-in-Differences (DDD) Model. We construct a DDD model to account for the unobservable time-varying characteristics of physicians. Specifically, we focus on the set of physicians that were active on both WeDoctor and Haodf during our study period and compare their performance in individual consultations across the two platforms. Our DDD model is given as follows:

$$\Delta Y_{it} = \Delta \beta_0 + \beta_1 After_{it} + \eta \Delta Z_{it} + \Delta \theta_t + \Delta \sigma_i + \varepsilon_{it}. \quad (2)$$

Here, $\Delta Y_{it} (= Y_{it}^W - Y_{it}^H)$ represents the difference in the number of satisfied reviews for physician i on day

⁵ While our observations are at the physician-day level, physicians often encounter external shocks that could influence their consultation on OMC platforms. In practice, these shocks often exhibit seasonality at the month level. As such, we follow previous research (e.g., Vana et al. 2018, Yue et al. 2019, Gutiérrez et al. 2020) to focus on time-fixed effects at the month level. We also replicate our analyses at the week level, and the results are consistent.

t between the two platforms (i.e., $SatiNum_{it}^W - SatiNum_{it}^H$). For simplicity, we use superscripts W and H to denote measurements related to WeDoctor and Haodf, respectively. β_1 is our DDD estimator. It captures the impact of team participation on physicians' performance in individual consultations. ΔZ_{it} ($=Z_{it}^W - Z_{it}^H$) is a vector of control variables that include $\Delta SatiCumNum_{i,t(-1)}$, which captures the cumulative number of satisfied reviews physician i received before day t on WeDoctor minus that of Haodf, $\Delta PatientNum_{i,t(-10)}$, which captures the number of individual patients of physician i over the last 10 days before day t on WeDoctor minus that of Haodf, $\Delta AvgResponseLen_{i,t(-10)}$, which captures the average length of physician i 's responses over the last 10 days before day t on WeDoctor minus that of Haodf, and $Tenure_{i,t(-1)}$. We construct the control variables $\Delta PatientNum_{i,t(-10)}$ and $\Delta AvgResponseLen_{i,t(-10)}$ using a lead time (e.g., 10 days), because, in practice, it often takes several days before patients review a physician's consultation service.⁶ $Tenure_{i,t(-1)}$ captures the tenure (measured by day) of physician i on WeDoctor. We also include the physician-fixed effects $\Delta \sigma_i$ and time-fixed effects $\Delta \theta_t$.

Coarsened Exact Matching. While the DDD model in Equation (2) alleviates the concern over the unobserved physician-specific time-varying factors, it is still possible that physicians who chose to join a team on WeDoctor are inherently different from those who did not. In light of this, we use coarsened exact matching (CEM) to identify similar physicians in the treatment and control groups. Specifically, we focus on two sets of variables that affect a physician's decision to join a consultation team on WeDoctor: the physician's profession-related characteristics (e.g., professional title/rank and whether affiliated with elite hospitals) and contributions (i.e., the number of individual patients, the number of satisfied reviews, and the average length of responses) across the two platforms in the first month after registering on WeDoctor. The former captures the physician's skills, abilities, and experience, which in turn play roles in the decisions to join a consultation team (Li and van Knippenberg 2021), and the latter captures the physician's engagement level on WeDoctor.

⁶ We also replicate the analyses with other lead time options (i.e., 7 days and 14 days), and the results remain qualitatively unchanged across these variations.

We use the 2,670 physicians who were active on the two platforms during our study period as the matching pool. After CEM, we are left with a sample of 338 matched physicians, where 169 physicians participated in team consultation on WeDoctor, and 169 did not. Appendix A presents the details of the CEM procedure and the balance check. Overall, we do not find significant differences between the two groups across all the matching variables. Table 2 summarizes the descriptive statistics of key variables.

Table 2. Descriptive Statistics of Key Variables

Variable	Description	Mean	Std. Dev.	Min	Max
$\Delta SatiNum_{it}$	The number of satisfied reviews physician i received on day t on WeDoctor minus that of Haodf	0.008	0.482	-30	89
$After_{it}$	A binary variable indicating whether physician i was part of a consultation team on day t on WeDoctor	0.315	0.465	0	1
$\Delta SatiCumNum_{i,t(-1)}$	The number of satisfied reviews physician i received before day t on WeDoctor minus that of Haodf	3.726	83.26	-1,441	1085
$\Delta PatientNum_{i,t(-10)}$	The number of individual patients of physician i over the last 10 days before day t on WeDoctor minus that of Haodf	-0.816	6.371	-484	253
$\Delta AvgResponseLen_{i,t(-10)}$	The average length of physician i 's responses over the last 10 days before day t on WeDoctor minus that of Haodf	2,416	61,784	-77,375	1.01e+07
$Tenure_{i,t(-1)}$	The number of days since physician i registered on WeDoctor before day t	673.7	520.0	0	2654

Notes: The total number of observations is 342,586 at the physician-day level.

4.2. Results

Main Results. Table 3 presents the estimation results from the DDD model specified in Equation (2). The DDD coefficient, $After_{it}$, is positive and significant (β_1 is 0.042 and p -value = 0.003), suggesting that compared to their individual consultations on Haodf, physicians who joined consultation teams on WeDoctor received more satisfied reviews for their individual consultations on WeDoctor (after the team participation) than those who did not. Given the average difference in the number of satisfied reviews between the two platforms is 0.008 (as shown in Table 2), the observed increase of 0.042 in the difference is also meaningful from a practical point of view. In particular, this result provides strong evidence that team participation has a significant, positive impact on physicians' performance in individual consultations.

Table 3. The Impact of Team Participation on Performance in Individual Consultation

Dependent Variable:	(1)
<i>After</i>	0.042*** (0.014)
<i>ASatiCumNum</i>	0.002 (0.001)
<i>APatientNum</i>	0.011** (0.005)
<i>AAvgResponseLen</i>	-2.52e-08 (2.08e-07)
<i>Tenure</i>	8.20e-05 (7.32e-05)
Time FEs	YES
Physician FEs	YES
Adjusted R ²	0.092
Observations	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

Robustness Checks. To validate our main findings, we conduct a series of robustness checks.

Table 4 summarizes the specific robustness concerns we address along with the respective validation strategies and the results.

Table 4. Summary of Robustness Checks

Robustness Concerns and References	Analyses and Findings
Concerns about the unobservable physician-specific time-varying factors (Appendix B)	(i) a DD model on physicians' individual consultations on WeDoctor (ii) a DD model on physicians' individual consultations on Haodf (iii) a relative time model
Parallel-Trend assumption (Table 5 and Appendix C)	(iv) a placebo test Finding: the increase in the number of satisfied reviews only happened on WeDoctor, not Haodf, and the parallel-trend assumption is not violated.
Selection concerns regarding team formation (Appendix D)	(i) a hurdle model (ii) a look-ahead approach (iii) an instrumental variable approach Finding: the results are consistent with those in Table 3
Incorrect functional form of estimator (Appendix E)	(i) a model with an alternative dependent variable (ii) a model with additional control variables (iii) models with alternative time window (iv) a subsample approach and a bootstrapping method Finding: the results are consistent with those in Table 3
Incorrect form of matching (Appendix F)	(i) CEM with alternative specifications (ii) propensity score matching (PSM) with or without replacement (iii) no matching Finding: the results are consistent with those in Table 3

Unobservable Physician-Specific Time-Varying Factors. As explained in Section 4.1, the DDD model specified in Equation (2) allows us to account for physician-specific time-varying factors more effectively than the DD model in Equation (1). As a robustness check, we conduct two DD estimations to investigate the potential impact of unobservable physician-specific time-varying factors. The rationale is that if there exist any unobservable physician-specific time-varying factors that affect both physicians' team participation decision and their performance in individual consultations, we should expect to see changes in the number of satisfied reviews of physicians' individual consultations on Haodf, albeit the absence of the team consultation feature. For example, if it is the increase of domain knowledge or work experience that drives physicians' team joining and performance improvement in individual consultations on WeDoctor, we should see an increased number of satisfied reviews of these physicians on Haodf. Similarly, if it is the personal aspiration (e.g., the willingness to learn new knowledge) that motivates physicians' team joining and performance improvement on WeDoctor, we should observe performance improvement in individual consultations on Haodf as well. Based on the estimation results in Tables B.1 and B.2 in Appendix B, physicians who joined consultation teams received more satisfied reviews for their individual consultations on WeDoctor but not on Haodf, suggesting that the physician-specific time-varying factors are unlikely to affect the relationship between physicians' team participation and their performance in individual consultations.

Parallel-Trend Assumption. The validity of the DDD approach rests on the parallel trend assumption. That is, prior to joining consultation teams, physicians in the treatment group should not significantly differ from their counterparts in the control group in terms of the difference in $SatiNum_{it}$. Following prior literature (Jung et al. 2019, Hou et al. 2024), we use a leads-lags model and extend Equation (2) by adding a series of time dummies that indicate different chronological distances relative to the time when physician i joined a team:

$$\Delta Y_{it} = \gamma_0 + \sum_{\tau=t-4}^{t+3} \gamma_{\tau}(After_{i\tau}) + \eta \Delta Z_{it} + \Delta \sigma_i + \Delta \theta_t + \varepsilon_{it}. \quad (3)$$

Similar to Equation (2), the dependent variable ΔY_{it} represents the difference in the performance in

individual consultations between the two platforms. $After_{it}$ denotes the time dummies (i.e., $After_{it} = 1$ if physician i joined a consultation team at time τ). To avoid the dummy variable trap (Jung et al. 2019), we use $t-1$ as the baseline and exclude it from the estimation. Table 5 presents the estimation results. We can observe that all coefficients corresponding to pre-treatment periods are not significant (p -values > 0.1), whereas those corresponding to the post-treatment periods are significant (p -values < 0.05), suggesting that there are no systematic differences between the treatment and control groups in the pre-treatment periods and the parallel assumption is not violated.

Table 5. A Relative-Time Model

Dependent Variable:	(1) $\Delta SatiNum$
Relative time ($t - 4$)	-0.025 (0.022)
Relative time ($t - 3$)	0.009 (0.014)
Relative time ($t - 2$)	0.002 (0.011)
Relative time ($t + 0$)	0.040*** (0.015)
Relative time ($t + 1$)	0.038*** (0.014)
Relative time ($t + 2$)	0.029** (0.016)
Relative time ($t + 3$)	0.048*** (0.016)
Control Variables	YES
Time FEs	YES
Physician FEs	YES
Adjusted R ²	0.092
Observations	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

We also illustrate the estimation results of the relative-time model in Figure 5. Here, the x-axis denotes different time periods, the y-axis shows the estimated coefficients of γ_j , and the dashed lines represent the 95% confidence intervals. Again, the performance differences are insignificant during the pre-treatment periods (i.e., $t-2$, $t-3$, and $t-4$) but significant during the post-treatment periods (i.e., t , $t+1$, $t+2$, and $t+3$).

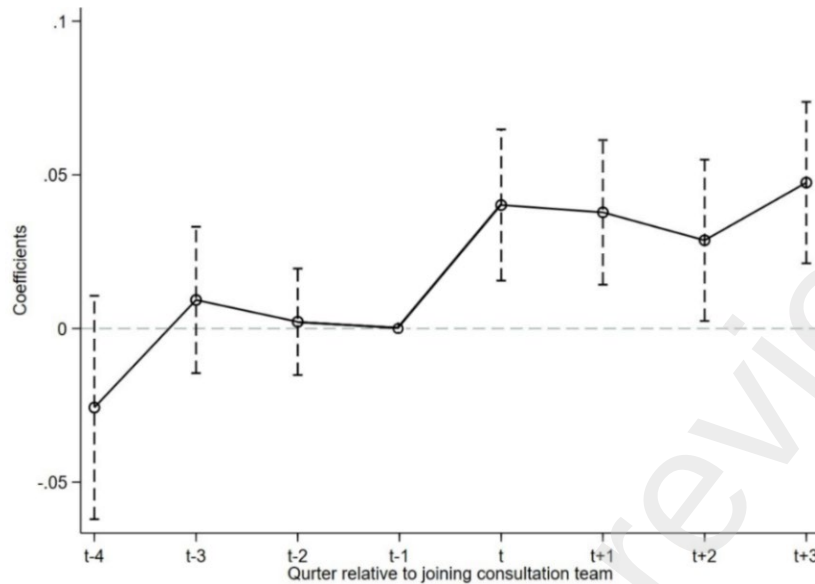


Figure 5. Illustration of Test Results of the Parallel-Trend Assumption

Further, we conduct a placebo test to verify the parallel-trend assumption. To do so, we first replicate the DDD estimation with a shorter time window (i.e., two quarters before and after the date a physician joined a consultation team). It can be seen from Column 1 of Table C.1 in Appendix C that the DDD estimate remains positive and significant. Next, for all matched physicians, we set their adoption date of team consultation to one quarter earlier, that is, $t-1$ corresponds to the post-treatment periods, whereas $t-2$, $t-3$, and $t-4$ correspond to the pre-treatment periods. We then re-estimate Equation (2) with this modified time window. Column 2 of Table C.1 shows that the DDD estimate is not significant (-0.013 , $p\text{-value} > 0.1$), which reinforces that there are no significant differences between physicians in the treatment and control groups prior to team participation.

Self-Selection Concern. Following Morgan (2015), we conduct additional tests to account for selection issues related to previous performance and expectations of future performance. To begin with, since our matching strategy already accounts for physicians' pre-treatment performance that could influence their team participation decision, we focus on confounding factors that may affect the likelihood of a physician being accepted into a team by other team members and quantify the effects through a hurdle model (Cragg 1971). The hurdle model consists of a binary choice component that accounts for whether or

not a physician is accepted into a team, which may depend on physicians' credentials and affiliations as well as their previous experience on WeDoctor (e.g., the number of patients over the last 10 days and the tenure on WeDoctor), and an additional component that captures the impact of team participation on individual consultations. The estimation results from the hurdle model (see Table D.1 in Appendix D) are largely consistent with those in Table 3.

We employ two strategies to address self-selection issues related to expectations of future performance. First, we focus on physicians in the treatment group and conduct a look-ahead analysis. The rationale is that physicians who eventually join consultation teams should share similar characteristics and thereby hold similar expectations of future benefits from joining a team (Huang et al. 2021a). To operationalize this analysis, we use half of the treated physicians who adopted team consultation earlier as the treatment group and the other half as the control group. The results of this look-ahead analysis are largely consistent with those in Table 3 (see Table D.2 in Appendix D for details).

Second, we apply an instrumental-variable approach using two-stage least squares (2SLS) estimation. We use the ratio of physicians who join consultation teams within the same hospital (i.e., $RatioHospital_{it}$) and the ratio of physicians who join consultation teams from the same department (i.e., $RatioDepartment_{it}$) as instrumental variables. The rationale is that the adoption rate of team participation among peers from the same unit (e.g., hospital or department) should positively correlate with a physician's participation decision (Zhang et al. 2018a) but not directly affect the physician's performance on individual consultations or the expectations of achieving good performance through joining a team. We conduct a set of tests to examine the relevance and strength of the instrumental variables, including the Cragg-Donald Wald F statistic (Stock and Yogo 2002), the Kleibergen-Paap rk Lagrange multiplier statistics (Kleibergen and Paap 2006), and Sargan-Hansen test for overidentification. Table D.4 reports the results, which are consistent with those from Table 3. To address the concern of forbidden regression, we also follow the three-stage procedure recommended by Adams et al. (2009) and Angrist and Pischke (2009) and replicate the 2SLS estimation. The results remain robust (see Table D.6).

Alternative Model Specifications and Matching Strategies. We replicate the DDD estimation

with alternative model specifications and matching strategies. To begin with, we consider an alternative dependent variable, $\Delta SatiRatio_{it}$, which captures the difference in the proportion of satisfied patients for physician i during the last 10 days before day t between the two platforms, i.e., $\Delta SatiRatio_{it} = SatiRatio_{it}^W - SatiRatio_{it}^H$. Column 1 of Table E.1 in Appendix E shows that the estimation results remain qualitatively the same as those in Table 3. We then add additional control variables, including the differences in the cumulative number of individual patients served, the total expected earnings from individual consultations, and the total number of questions answered to further account for physician heterogeneity. The results in Table E.2 in Appendix E are consistent with those in Table 3. We also perform the DDD estimation using different time windows—30, 60, 90, 120, and 150 days—before and after joining consultation teams. This helps ensure that the observed effects are persistent. The results shown in Table E.3 in Appendix E indicate that the main findings from Table 3 remain unchanged. Further, given the limited sample size after CEM (i.e., 338), we show the stability of our estimation in two ways. First, we randomly select 150 physicians (half from the treatment group and half from the control group) and replicate the DDD estimation. Second, we use bootstrapping to repeat the DDD estimation process 10,000 times (Pu et al. 2022). The results in Table E.4 in Appendix E are consistent with those in Table 3.

We also consider alternative matching strategies. Recall that in the main analysis, we perform CEM by restricting the same number of physicians in the treatment and control groups. As a robustness check, we first conduct another CEM without such a restriction (Blackwell et al. 2009). Next, we use additional matching variables for CEM, including the rank of the city (based on its business attractiveness) where a physician's affiliated hospital is located, the physician's gender, and the medical department that the physician serves. Finally, we conduct one-to-one PSM without replacement (i.e., physicians in the control group cannot be matched with more than one physician in the treatment group) and with replacement (i.e., physicians in the control group can be matched with more than one physician in the treatment group) and re-estimate Equation (2) accordingly. Table F.1 reports the estimation results under the alternative matching strategies. To facilitate comparison, we also include the estimation results using the full sample without matching (Column 5 of Table F.1). Overall, we find that the results are consistent with those in Table 3.

5. Mechanism Exploration

So far, we have examined the impact of team participation on physicians' performance in individual consultations. In this section, we explore potential mechanisms and discuss the extent to which they are aligned with the empirical evidence. Drawing upon the literature, we first consider peer effects (Section 5.1) and knowledge sharing (Section 5.2), and then turn to other alternative explanations (Section 5.3).

5.1. Peer Effects

A salient feature of teamwork is that individual team members are subject to peer effects (Manski 1993). To test the possible mechanism route of peer effects, we adopt the social comparison perspective (Smith 2000) and examine how peer effects can manifest among physicians who join consultation teams. Specifically, we first investigate whether the increased performance in individual consultations is from the enhanced effort and then explore under what conditions social comparison becomes more pronounced.

Physician Effort in Individual Consultation. Prior studies have shown that relative performance feedback can trigger social comparison, which in turn drives workers to increase their effort and work harder (e.g., Mas and Moretti 2009, Song et al. 2018, Ye et al. 2022). In our case, physicians working in the same team can observe the number of individual consultations and the number of satisfied reviews for themselves and their peers. Such publicly disclosed information constitutes a form of relative performance feedback, which motivates physicians, via social comparison (Roels and Su 2014), to exert more effort in addressing patients' needs.

Following this rationale, we compare a physician's effort before and after joining a consultation team. To quantify a physician's effort in individual consultations, we focus on the text features of physicians' replies to patients' inquiries. We first consider the length (i.e., the number of Chinese characters) of physicians' replies, as longer replies generally require more effort. We replace the dependent variable in Equation (2) with the difference in the length of replies between the two platforms for physician i at time t and replicate the DDD estimation. According to Column 1 of Table 6, the length of physicians' replies increased significantly after joining consultation teams.

Table 6. Physician Effort on Individual Consultation

	(1)	(2)	(3)	(4)
Dependent Variable:	$\Delta ResponseLen$	$\Delta Emotion$	$\Delta NewMedical$	$\Delta Entropy$
<i>After</i>	81.158* (46.210)	0.048** (0.020)	0.014*** (0.004)	0.047** (0.019)
$\Delta SatiCumNum$	-0.176 (0.289)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
$\Delta PatientNum$	10.666*** (2.324)	0.021*** (0.007)	0.004*** (0.001)	0.038*** (0.008)
$\Delta AvgResponseLen$	-0.002 (0.009)	-5.23e-07 (4.11e-07)	-1.08e-07 (1.45e-07)	-6.92e-08 (5.20e-07)
<i>Tenure</i>	0.019 (0.217)	0.000 (0.000)	0.000* (0.000)	0.000 (0.000)
Time FEs	YES	YES	YES	YES
Physician FEs	YES	YES	YES	YES
Adjusted R ²	0.097	0.039	0.015	0.162
Observations	152,082	152,082	152,082	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses. We also check whether joining consultation teams increases the time that physicians spend on the platform. Specifically, we use the difference in the physicians' average response time across the two platforms, i.e., $\Delta ResponseTime_{it}$, as the dependent variable and rerun Equation (2). The results show that the response time of physicians does not significantly change after team participation.

Next, we turn to the content of physicians' replies and examine the differences in three effort indicators between the two platforms: (1) the average number of emotional-support terms used per day (i.e., $\Delta Emotion_{it}$), (2) the average number of new medical terms used per day (i.e., $\Delta NewMedical_{it}$), and (3) the diversity of topics per day (i.e., $\Delta Entropy_{it}$). For physician i 's replies on day t , we first perform word segmentation, and then identify the emotional-support terms ($\Delta Emotion_{it}$) and medical terms by matching the words in existing dictionaries.⁷ This procedure has been used in prior studies on digital platforms (Goes et al. 2014, He et al. 2020). After identifying the medical terms, we calculate $\Delta NewMedical_{it}$ by comparing the medical terms physician i used on day t with the medical terms used before day t . To determine the diversity of topics used in physicians' replies, we follow a similar procedure used by Bapna et al. (2019). Specifically, we first apply latent Dirichlet allocation (LDA) to extract topics from physicians' replies and then calculate Shannon entropy. The input of LDA is physicians' replies to individual patients on the two

⁷ The dictionary used for medical terms is from Peking University Open Research Data (<https://opendata.pku.edu.cn/>), and the dictionary employed for emotional-support terms is from Linguistic Inquiry and Word Count (<https://www.liwc.app/>).

platforms, and the output is the predefined number (K) of topics from physicians' replies and a posterior distribution over all K topics for each reply. Specifically, for each reply j , the LDA provides the weight of reply j associated with each topic with a K -vector, $\langle T_{j,1}, T_{j,1}, \dots, T_{j,k} \rangle$, where $0 < T_{j,k} < 1$ and $\sum_{k=1}^K T_{j,k} = 1$. The greater $T_{j,k}$ means the reply j is more likely associated with topic k . We test the models with $K = 10, 15$, and 20 topics and find that 10 is the optimal number of latent topics in terms of posterior log-marginal likelihood (Griffiths and Steyvers 2004, Bapna et al. 2019). After obtaining the topic distribution of physicians' replies, we calculate the difference in the average topics of physician i 's replies on day t to measure $\Delta Entropy_{it}$.

We replace the dependent variable in Equation (2) with $\Delta Emotion_{it}$, $\Delta NewMedical_{it}$, and $\Delta Entropy_{it}$, and replicate the DDD estimation accordingly. The estimation results (see Columns 2-4 of Table 6) provide strong evidence that physicians spend more effort replying to individual consultations after joining a team.⁸ We also replicate the analysis using these variables by running separate DD estimations and present the results in Table 7. The estimates for $After_{it}$ are significant for the effort indicators of physicians' individual consultations on WeDoctor (Columns 1-4) but not on Haodf (Columns 5-8), confirming that the increased effort associated with the adoption of team consultation only manifested on the platform where physicians actually involved in team consultations.

Differential Effects of Social Comparison. So far, we have provided empirical evidence of peer effects via social comparison. In practice, the magnitude/strength of the social-comparison-induced motivating effects typically varies in the composition of the peer groups (Mas and Moretti 2009, Amodio and Martinez-Carrasco 2018). Therefore, we conduct a set of heterogeneity analyses. Specifically, we are interested in the impact of the following factors: visibility of peers (i.e., teammates), peers' performance in individual consultations, social relationship between a focal physician and his peers, and the focal physician's characteristics (e.g., recognition in the field, past performance in individual consultations).

⁸ We also conducted semi-structured interviews with a group of physicians who have been actively involved in team consultations. The majority of the physicians mentioned that their teammates' performance had an impact on their own performance in individual consultations.

Table 7. Physician Effort on Individual Consultation on WeDoctor and Haodf

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
D.V.:	<i>ResponseLen^W</i>	<i>Entropy^W</i>	<i>NewMedical^W</i>	<i>Emotion^W</i>	<i>ResponseLen^H</i>	<i>Entropy^H</i>	<i>NewMedical^H</i>	<i>Emotion^H</i>
<i>After</i>	82.884* (45.523)	0.031** (0.012)	0.006*** (0.002)	0.063*** (0.019)	-0.171 (0.541)	-0.014 (0.012)	-0.004 (0.002)	-0.045 (0.034)
<i>SatiCumNum</i>	-0.415 (0.255)	0.000 (0.000)	0.000 (0.000)	-0.001 (0.001)	0.174 (0.113)	0.000 (0.000)	0.000** (0.000)	0.000 (0.001)
<i>PatientNum</i>	12.368** (5.895)	0.047*** (0.009)	0.005*** (0.001)	0.054*** (0.010)	3.041 (1.861)	0.023** (0.010)	0.001 (0.001)	0.035 (0.023)
<i>AvgResponseLen</i>	-0.002 (0.009)	-3.25e-07 (2.60e-07)	-8.64e-08** (3.50e-08)	-0.000*** (3.68e-07)	0.049*** (0.012)	0.000 (0.000)	8.67e-06 (5.40e-06)	0.000 (0.000)
<i>Tenure</i>	0.002 (0.218)	0.000 (0.000)	-8.35e-06 (0.000)	0.000 (0.000)	-0.032*** (0.010)	0.000 (0.000)	0.000* (0.000)	0.000 (0.000)
Time FEs	YES	YES	YES	YES	YES	YES	YES	YES
Physician FEs	YES	YES	YES	YES	YES	YES	YES	YES
Adjusted R ²	0.097	0.137	0.037	0.064	0.331	0.262	0.034	0.168
Observations	152,082	152,082	152,082	152,082	152,082	152,082	152,082	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses. The control variables in Columns 1-4 describe results for WeDoctor and Columns 5-8 for Haodf.

To begin with, previous research has shown that visibility of peers' behavior can lead to strong social comparison (Bollinger et al. 2022), which in turn drives behavioral improvement (Baldwin and Mussweiler 2018). In our context, both the structured design and transparency offered by the virtual team setting on OMC platforms make it easier for physicians to track their peers' performance over time relative to that in a traditional offline setting. In particular, as illustrated in Figure 3, a virtual team's description includes a list of team members with explicit rankings (orders). These rankings are typically correlated with a member's tenure in the team (i.e., members who joined a team earlier would be ranked higher) and do not change over time. Physicians with a high ranking on the page are more visible to other members of the team (Collins et al. 2018), and are hence more likely to be observed and used as comparison targets. This is particularly the case on the mobile app interface:⁹ Physicians ranked in the top three are displayed on the first screen and the rest members are displayed on the subsequent screens. In light of this, we use a physician's virtual ranking on the team description page as a proxy for his visibility. If social comparison is at play, the performance improvement of physicians with rankings below the top three would be more

⁹ In recent years, patient-physician interactions have increasingly moved from the web to mobile devices, see <https://www.iimedia.cn/c1020/75792.html>, accessed on January 10, 2025.

pronounced than their peers ranked in the top three.

Following this rationale, we create a binary variable $NonTop3_i$, which indicates whether physician i has a ranking below the top three on the team description page (i.e., a less prominent page position). Using this variable as a moderator in Equation (2), we re-estimate the DDD model and present the results in Column 1 of Table 8. We can observe that the coefficient of the interaction term is positive and significant, indicating that the performance improvement is indeed more pronounced for physicians with lower rankings relative to the ones with top rankings.

Table 8. Moderating Effect of Rankings of Members on the Team Description Page

Sample	Subsample	Rematched Sample with Balanced Observables
	(1)	(2)
Dependent Variable:	$\Delta SatiNum$	$\Delta SatiNum$
<i>After</i>	0.017 (0.014)	-0.029* (0.017)
<i>After</i> × <i>NonTop3</i>	0.050* (0.030)	0.068** (0.033)
$\Delta SatiCumNum$	0.002* (0.001)	0.004*** (0.001)
$\Delta PatientNum$	0.011** (0.005)	0.007 (0.006)
$\Delta AvgResponseLen$	-2.55e-08 (2.20e-07)	-1.57e-05 (3.01e-05)
<i>Tenure</i>	8.17e-05 (7.34e-05)	0.000 (0.000)
Time FEs	YES	YES
Physician FEs	YES	YES
Adjusted R ²	0.093	0.199
Observations	152,082	48,154

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

To account for the possibility that the observed heterogeneous effects are due to the difference in physicians' characteristics (e.g., expertise or experience) other than the rankings (orders) on the team description page, we further create a matched sample using the 106 physicians with top three rankings and the 63 physicians with lower rankings using a CEM procedure similar to the one outlined in Appendix A. The resulting sample consists of 110 physicians. Using this sample, we re-estimate the moderating effect of $NonTop3_i$. The results (see Column 2 of Table 8) are qualitatively unchanged compared to those in Column 1.

Next, we explore to what extent peers' performance in individual consultations affects the materialization of social comparison. In particular, prior research has suggested that upward comparison can induce performance improvement (Festinger 1954). If the performance improvement is primarily driven by social comparison, we would expect physicians to exert more effort when their teammates have received better ratings and/or more satisfied reviews. Following this rationale, we create two variables, $\Delta PeerIndividualPatient_{i,t(-10)}$ and $\Delta PeerIndividualSatisfaction_{i,t(-10)}$, to capture the difference in the average number of individual consultations served by physician i 's peers (i.e., teammates) and the difference in the average number of satisfied reviews received by the physician i 's peers over the last 10 days before day t between WeDoctor and Haodf. We include these two variables as moderators in Equation (2) and rerun the DDD estimation. As shown in Table 9, the coefficients of the interaction terms are positive and significant, suggesting that the performance of peers in individual consultations can moderate the spillover effect of team participation, thereby further corroborating the social comparison explanation.

Table 9. Moderating Effect of Peer Performance

	(1)	(2)
Dependent Variable:	$\Delta SatiNum$	$\Delta SatiNum$
<i>After</i>	0.043*** (0.014)	0.043*** (0.014)
<i>After</i> × $\Delta PeerIndividualPatient$	4.51e-05*** (1.72e-05)	-
$\Delta PeerIndividualPatient$	-2.31e-06 (1.74e-06)	-
<i>After</i> × $\Delta PeerIndividualSatisfaction$	-	6.34e-05** (2.48e-05)
$\Delta PeerIndividualSatisfaction$	-	-2.22e-06 (2.02e-06)
$\Delta SatiCumNum$	0.002 (0.001)	0.002 (0.001)
$\Delta PatientNum$	0.011** (0.005)	0.011** (0.005)
$\Delta AvgResponseLen$	-2.54e-08 (2.08e-07)	-2.52e-08 (2.08e-07)
<i>Tenure</i>	0.000 (0.000)	0.000 (0.000)
Time FEs	YES	YES
Physician FEs	YES	YES
Adjusted R ²	0.092	0.092
Observations	152,082	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

Drawing upon prior research about the tie strength and social comparison among peers (e.g., Lee and Griffith 2019), we further delve into the roles of the relationship between peers and focal physicians on the materialization of social comparison. Specifically, we examine how the impact of team participation on individual performance varies in geographical proximity between team members. We first looked at whether the presence of colleagues from the same hospital in a team would moderate the impact of team participation. We created a dummy variable, *SameHospitalPeer_i*, to capture whether physician *i* has at least one teammate from the same hospital and incorporate it into Equation (2) as a moderator. The results are presented in Column 1 of Table 10.

Table 10. Moderating Effect of Team Members' Geographical Proximity

	(1)	(2)
Dependent Variable:	$\Delta SatiNum$	$\Delta SatiNum$
<i>After</i>	-0.006 (0.021)	-0.015 (0.035)
<i>After</i> × <i>SameHospitalPeer</i>	0.052** (0.024)	-
<i>After</i> × <i>SameCityPeer</i>	-	0.059* (0.035)
$\Delta SatiCumNum$	0.002 (0.001)	0.002 (0.001)
$\Delta PatientNum$	0.011** (0.005)	0.011** (0.005)
$\Delta AvgResponseLen$	-3.01e-08 (2.10e-07)	-2.66e-08 (2.09e-07)
<i>Tenure</i>	0.000 (0.000)	0.000 (0.000)
Time FEs	YES	YES
Physician FEs	YES	YES
Adjusted R ²	0.092	0.092
Observations	152,082	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses. The estimations of *SameHospitalPeer* and *SameCityPeer* are absorbed by the Physician FEs.

We can observe that the coefficient of the interaction term is positive and significant (0.052, p -value = 0.028). This finding indicates that the positive spillover effect of team participation on individual consultations is stronger for those who have teammates from the same hospital. We then replicated the analysis by considering physicians with teammates from the same city by introducing a dummy variable—*SameCityPeer_i*. The results (see Column 2 of Table 10) indicate that having teammates from the same city also amplifies the impact of team participation. These findings resonate with the insights from Zhang et al.

(2018b), highlighting the nuanced effects of social comparison among peers with strong ties.

Finally, we also look into how physician characteristics affect the materialization of social comparison. For example, renowned physicians (e.g., physicians from elite hospitals or those considered to be domain experts) may react to peer pressure differently from those who are less well-known. To investigate whether the effects of social comparison vary in focal physicians' characteristics, we first create two dummy variables to capture whether a physician is from an elite hospital (i.e., *EliteHospital_i*) or is a domain expert (i.e., *Expert_i*). We replicate the DDD estimation by including the corresponding interaction terms in Equation (2). It follows from Columns 1-2 of Table 11 that the coefficients of these two interaction terms are not significant.

Table 11. Moderating Effect of Focal Physicians' Characteristics

Dependent Variable:	(1)	(2)	(3)	(4)
	$\Delta SatiNum$	$\Delta SatiNum$	$\Delta SatiNum$	$\Delta SatiNum$
<i>After</i>	0.002 (0.099)	0.067*** (0.025)	0.046*** (0.014)	0.059*** (0.015)
<i>After</i> × <i>EliteHospital</i>	0.040 (0.100)	-	-	-
<i>After</i> × <i>Expert</i>	-	-0.049 (0.033)	-	-
<i>After</i> × <i>PreIndividualPatient</i>	-	-	-5.76e-05*** (8.76e-06)	-
<i>After</i> × <i>PreIndividualSatisfaction</i>	-	-	-	-0.001** (0.000)
$\Delta SatiCumNum$	0.002 (0.001)	0.002 (0.001)	0.002 (0.001)	0.002* (0.001)
$\Delta PatientNum$	0.011** (0.005)	0.011** (0.005)	0.011** (0.005)	0.011** (0.005)
$\Delta AvgResponseLen$	-2.58e-08 (2.08e-07)	-2.12e-08 (1.97e-07)	-2.62e-08 (2.08e-07)	-3.86e-08 (2.10e-07)
<i>Tenure</i>	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Time FEs	YES	YES	YES	YES
Physician FEs	YES	YES	YES	YES
Adjusted R ²	0.092	0.093	0.093	0.093
Observations	152,082	152,082	152,082	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses. The estimations of *EliteHospital*, *Expert*, *PreIndividualPatient*, and *PreIndividualSatisfaction* are absorbed by the Physician FEs.

In a similar vein, we also investigate whether the effects of social comparison vary in focal physicians' performance in individual consultation before joining a team. To do so, we create two variables *PreIndividualPatient_i* and *PreIndividualSatisfaction_i* to capture the number of patients served by physician

i and the overall patient satisfaction level of i prior to joining a team. Columns 3 and 4 of Table 11 reveal that the coefficients of the two interaction terms of interest are both negative and significant, suggesting that the effect of social comparison is more pronounced for lower performers in individual consultations before joining a team. Taken together, the results presented in Table 11 echo the findings from Cassar and Ko (2023) that peer effects are weaker for individuals who have superior individual performance.

5.2. Knowledge Sharing

The positive impact of team participation on subsequent performance in individual tasks may also be attributed to knowledge sharing in a team (Chan et al. 2014, Song et al. 2018). That is, physicians acquire knowledge through observing and interacting with their teammates. Despite the theoretical plausibility, however, this mechanism is unlikely to be the main driver of the positive effect of team participation shown in Table 3. In fact, our DDD approach already accounts for physician-specific factors that are subject to change under knowledge sharing (e.g., domain knowledge and work experience): If the observed effect of team participation were primarily driven by knowledge sharing, we would expect that physicians' performance improvement on WeDoctor paralleled that on Haodf, i.e., the estimate of β_1 in Equation (2) should have been insignificant and the estimate of β_1 in Equation (B.1) should have been significant, because it is unlikely that physicians would apply the knowledge and/or experience acquired through their peers exclusively on one platform (WeDoctor).¹⁰

We conduct two additional sets of tests to explore to what extent knowledge sharing contributes to the observed spillover effect of virtual teamwork. The first set of tests draws upon prior research on transactive memory systems (e.g., Kanawattanachai and Yoo 2007, Choi et al. 2010), which suggests that task-oriented communications are critical to effectively coordinating specialized knowledge among team members. If knowledge sharing plays a major role, the performance improvement should be more pronounced for physicians who have had more opportunities to interact with teammates directly (e.g.,

¹⁰ We also conduct a DD estimation by comparing team physicians' post-adoption performance across the two platforms and find that these physicians exhibited significantly better performance on WeDoctor than on Haodf, which is against the knowledge-sharing-based explanation.

collaboratively providing consultation) or indirectly (e.g., observing teammates' responses in team consultation sessions) immediately before day t (Wittenbaum and Park 2001). Following this rationale, we use the number of team-consultation patients served by physician i during the last 10 days before day t (i.e., $TeamPatient_{i,t(-10)}$) and the number of team-consultation patients served by teammates of physician i over the last 10 days before day t (i.e., $PeerTeamPatient_{i,t(-10)}$) as proxies for physician i 's within-team communication frequency. We include these two variables as moderators in Equation (2) and rerun the estimation. According to Table 12, the coefficients for the interaction terms are insignificant, suggesting that increased frequency of (direct and indirect) exposure to knowledge-sharing opportunities within a consultation team does not contribute to performance improvement in individual consultations.

Table 12. Impact of Within-team Communication Frequency on Individual Consultation

	(1)	(2)
Dependent Variable:	$\Delta SatiNum$	$\Delta SatiNum$
<i>After</i>	0.043*** (0.013)	0.041*** (0.014)
<i>After</i> × <i>TeamPatient</i>	-0.002 (0.004)	-
<i>TeamPatient</i>	0.002*** (0.000)	-
<i>After</i> × <i>PeerTeamPatient</i>	-	0.000 (0.001)
<i>PeerTeamPatient</i>	-	-0.001 (0.001)
$\Delta SatiCumNum$	0.002 (0.001)	0.002 (0.001)
$\Delta PatientNum$	0.011* (0.006)	0.011** (0.005)
$\Delta AvgResponseLen$	-2.69e-08 (2.07e-07)	-2.43e-08 (2.09e-07)
<i>Tenure</i>	0.000 (0.000)	0.000 (0.000)
Time FEs	YES	YES
Physician FEs	YES	YES
Adjusted R ²	0.092	0.092
Observations	152,082	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses. Since for physicians in the control group, $TeamPatient_{i,t(-10)}$ always takes the value of zero, we also replicate the estimation by focusing exclusively on physicians that adopted team consultation. The coefficients for the two interaction terms remain insignificant.

Overall, our analysis suggests that knowledge sharing, albeit its well-known role in enhancing individuals' performance in team settings, may not be the leading driver of the observed spillover effect. Nevertheless, we note that the above analyses do not rule out the possibility that team consultation can facilitate knowledge sharing but rather highlight the existence and importance of alternative mechanisms through which the positive spillover effect of team consultation emerges and persists in individual consultations.

5.3. Alternative Explanations

So far, our mechanism exploration has focused on the two most salient features that pertain to team settings: knowledge sharing and peer effects. In the rest of this section, we discuss other alternative explanations. First, it is possible that a patient's perception of a physician changes after the physician joins a team. For instance, serving as a team leader could enhance a physician's professional status, which may also increase the physician's credibility and trustworthiness. Such a Matthew effect (Azoulay et al. 2014) may lead to more satisfied reviews from individual patients for a team leader regardless of his effort exerted or actual performance in individual consultations. To test the plausibility of this explanation, we split the treated physicians into two groups based on whether the physician is a team leader or not and then re-estimated Equation (2). The rationale is as follows: If a status improvement plays a role, being a team leader should reinforce such an improvement and we would observe a stronger effect of team participation on performance improvement (which is measured by satisfied reviews) for team leaders relative to non-leaders. Table G.1 in Appendix G reports the results. Columns 1-2 show that the coefficient estimates of interest are positive and significant for both leaders and non-leaders in teams, and there is no significant difference¹¹ between the two coefficient estimates (t -value = 0.182). We also included an interaction term $After_{it} \times Leader_i$ in Equation (2). Column 3 shows that the coefficient estimate of the interaction term is not

¹¹ Since team leaders are typically among the top-ranked ones, such insignificant result seems to contradict the findings from Table 8. We conduct additional analyses by comparing the performance of leaders and non-leaders among physicians ranked in the top three of a team. Specifically, we re-estimate Equation (2) by including an interaction term $After_{it} \times Leader_i$. The coefficient estimate of this interaction term is insignificant (0.042, p -value > 0.1), suggesting that the leader role itself does not translate into the observed performance improvement, nor does it moderate the visibility effect shown in Table 8.

significant. These results jointly imply that the potential change in a patient's perception of a physician is not the leading explanation for the observed performance improvement in individual consultations.

Second, it is also possible that joining consultation teams may increase the time that a physician spends on WeDoctor but not on Haodf. If that is the case, the treated physicians are likely to provide more timely responses to individual patients, leading to a higher level of patient satisfaction (Zhang et al. 2019). To examine this possibility, we used the difference in physicians' average response time each day across the two platforms as the dependent variable and re-estimated Equation (2). The coefficient estimate of interest is insignificant (see Column 4 of Table G.1 in Appendix G for details), suggesting that there is no significant difference in the change of response time across the two platforms, and hence the observed performance improvement cannot, by large, be attributed to increased time physicians spent on WeDoctor after joining consultation teams.

To summarize, our mechanism exploration provides nuanced insights into how peer effects manifest in virtual teams and affect individual members. Specifically, our findings suggest that higher visibility of peers' performance is likely to drive team members to improve their individual performance beyond the team setting. Moreover, social comparison within teams intensifies when a physician has high-performing peers, shares strong ties with peers, and exhibits lower individual performance prior to joining a consultation team.

6. Discussions

In this paper, we study how participation and engagement in virtual teams affect members' performance in individual tasks. Using a rich data set from two leading OMC platforms, we find that team participation has a robust, positive spillover effect on physicians' performance in individual consultations. Further analysis suggests that such a positive spillover effect is likely driven by peer effects that manifest through social comparison, which, in turn, is moderated by the visibility of peers, peers' performance, tie strength between team members, and individual physicians' own performance.

Our study holds important implications for both academics and practitioners. To start with, our

study extends prior research on virtual teams. While researchers have made extensive progress in understanding the confluence of social, behavioral, and cognitive factors on the functioning of virtual teams, we are among the first to empirically investigate the impact of virtual teamwork on individual performance and explore the underlying mechanisms. In particular, our study demonstrates the effect of team participation on members' performance on individual tasks beyond the team setting, and the results enrich the literature by shedding new light on the boundary-spanning roles of virtual teams. Our mechanism exploration reveals that virtual teamwork can amplify the effects of social comparison by increasing the visibility of individuals' performance. As a result, it can motivate team members to exert more effort in their individual tasks and improve their individual performance outside the team setting. As organizations are increasingly adopting the hybrid workplace model, it is important to understand under what conditions the benefits of teamwork transcend the team setting and how to leverage those benefits. Our findings provide valuable insights for organizations as to how to harness the power of the boundary-spanning roles of virtual teams.

Our paper also provides much-needed guidance for the design and operation of OMC platforms. Specifically, while the rise of OMC platforms has significantly improved the connectivity among patients and physicians by reducing or even eliminating certain frictions in patient-physician encounters, there remain major challenges as to how to effectively match patients and physicians and facilitate value creation for both sides given the overall supply shortage of physicians. We empirically show that team consultation may complement rather than cannibalize individual consultation. In particular, our analysis indicates that joining consultation teams can motivate physicians to exert more effort in individual consultations, which helps increase patient satisfaction levels. Our findings from the mechanism exploration provide actionable insights for platform operators. For instance, encouraging lower-performing physicians to team up with high-performing colleagues may be beneficial to both groups of individuals when an efficient coordination mechanism is enabled.

To this point, we note that although our analysis has focused on the context of OMC platforms, our findings are generalizable to other platforms where individuals can choose to form teams on a voluntary

basis (e.g., crowdsourcing platforms, OSSD platforms, and MOOC platforms). Given the positive spillover effects of virtual teamwork, platforms should design better incentive schemes to encourage team formation and offer timely support to facilitate coordination and communication among team members. Moreover, to operationalize the coexistence of teamwork and individual work to the maximum benefit, platforms may consider providing relative performance feedback to team members, although such decision must be grounded based on a thorough evaluation of the user base.

In closing, we acknowledge that our study has several limitations, and there are many important open questions to be explored. For example, when investigating the impact of team participation on physicians' individual performance, we have not accounted for physicians' offline commitments or constraints. In practice, the team consultation experience may also affect physicians' in-person clinical services. Exploring such spillover effects from online channels to offline channels could offer more actionable insights into the integration of virtual care options into the existing care systems. Moreover, our empirical analysis is based on a sample of physicians who provide OMC services on two platforms during the study period, and we have not accounted for potential competition between these two platforms. However, as more OMC platforms start to offer team consultation services, it is likely that physicians (especially experts who specialize in certain areas) will join multiple platforms, which would generate different dynamics between team and individual consultations. Exploring the competitive landscape where different OMC platforms are bridged by physician teams will be an interesting direction for future research. Finally, while our mechanism exploration has focused on knowledge sharing and peer effects, we acknowledge that there may be other cognitive and/or behavioral factors that contribute to the positive spillover effects of team participation. Identifying the relevant factors and understanding the complex interplay of these factors will be a fruitful direction for future research.

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Team Makes You Better: Evidence from Online Medical Consultation Platforms

Online Appendices

Appendix A. Details of the Coarsened Exact Matching

We use coarsened exact matching (CEM) to match the physicians in the treatment group (i.e., physicians who participated in a consultation team) and those from the control group (i.e., physicians who did not participate in a consultation team) from the 2,670 physicians active on both WeDoctor and Haodf. We conduct the matching with two sets of matching variables (listed in Table A.1) that may influence physicians' decisions about whether to join consultation teams on WeDoctor. We include physicians' individual characteristics (professional title and whether they are from elite hospitals) and the differences in physicians' individual performance across the two platforms in the first month after registering on WeDoctor as matching variables.

Table A.1 Variables Used for Matching Physicians in the Treatment and Control Groups

Individual characteristics: -- <i>Profession title</i> : 1 for resident physician, 2 for attending physician, 3 for associate chief physician, and 4 for chief physician. -- <i>Elite hospital</i> : 1 indicates the physician is from a top (3A) hospital, and 0 otherwise.
Differences in individual performance across WeDoctor and Haodf (the first month after enrolling WeDoctor) -- <i>ΔNumber of individual patients</i> : The number of individual patients the physician replied to in the first month after registering on WeDoctor minus that of Haodf. -- <i>ΔNumber of satisfied reviews</i> : The number of satisfied reviews from individual patients the physician received in the first month after registering on WeDoctor minus that of Haodf. -- <i>ΔLength of physician responses</i> : The total number of words in physicians' responses in the first month after registering on WeDoctor minus that of Haodf.

We use CEM to ensure that we have the same number of physicians in the treatment and control groups. This is done by pruning observations within each stratum until there are the same numbers in the treatment and control groups (Iacus et al. 2012). Among the 292 physicians who joined consultation teams

and the 2,378 physicians who did not join any consultation teams, 338 physicians are matched. After matching, there are no significant differences between the two groups of physicians with respect to the two sets of matching variables (see Table A.2).

Table A.2 T-test of the Variables Before and After Matching

Variable	Sample	Mean		T-test	
		Treated	Control	t	$p > t $
Individual characteristics:					
Profession title	Unmatched	3.473	0.264	-56.01	0.000
	Matched	3.485	3.485	0.000	1.000
Elite Hospital	Unmatched	0.893	0.075	-59.25	0.000
	Matched	0.888	0.888	0.000	1.000
Differences in individual performance:					
Δ Number of individual patients	Unmatched	0.828	0.012	1.513	0.131
	Matched	-0.181	-0.967	-0.403	0.687
Δ Number of satisfied reviews	Unmatched	1.007	2.189	1.501	0.133
	Matched	-0.172	-0.308	-0.480	0.632
Δ Length of physician responses	Unmatched	9371	6614	-0.527	0.599
	Matched	2194	1769	-0.179	0.858

Appendix B. Estimation based on the Difference-in-Differences (DD) Approach

In this appendix, we first present the results of the DD model specified in Equation (1). It follows from Table B.1 that team participation has a positive effect (0.035, p -value < 0.001) on the satisfaction level of physicians' individual consultations on WeDoctor.

Table B.1 Robustness Checks with a DD Model on WeDoctor

Dependent Variable:	(1) <i>SatiNum^W</i>
<i>After</i>	0.035*** (0.010)
<i>SatiCumNum^W</i>	0.003*** (0.001)
<i>PatientNum^W</i>	-0.000 (0.000)
<i>AvgResponseLen^W</i>	-1.03e-08** (4.08e-09)
<i>Tenure^W</i>	-0.000 (0.000)
Time FEs	YES
Physician FEs	YES
Adjusted R ²	0.070
Observations	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses. *SatiCumNum^W* is the cumulative number of satisfied reviews physician i received before day t on WeDoctor, *PatientNum^W* is the number of individual patients of physician i over the last 10 days before day t on WeDoctor, *AvgResponseLen^W* is the average length of physician responses of physician i over the last 10 days before day t on WeDoctor, and *Tenure^W* is the number of days since physician i registers on WeDoctor before day t .

As explained in Section 4.1, a potential concern about the DD estimation using Equation (1) is that it does not control for unobservable physician-specific time-varying factors that may affect physicians' performance in individual consultations as well as their decisions to join a team. To alleviate the concern, we can run another DD estimation to examine how physicians' adoption of team consultation on WeDoctor affects their individual consultations on Haodf, where team consultation was not available during our study period. To do so, we consider the following model specification:

$$Y_{it}^H = \beta_0 + \beta_1 After_{it}^H + \eta Z_{it}^H + \theta_t^H + \sigma_i^H + \varepsilon_{it}^H, \quad (B.1)$$

where Y_{it}^H denotes the number of satisfied reviews physician i received on day t on Haodf (i.e., *SatiNum_{it}^H*).

The binary variable $After_{it}^H$ takes the same value as $After_{it}^W$ in Equation (1) for any given (i, t) combination.

Equation (B.1) is essentially a falsification test because the team consultation feature was not implemented on Haodf during the observation period. According to Table B.2, the impact of team participation on the satisfaction level of physicians' individual consultations on Haodf is not significant. Recall from Section 3.2 that the two platforms have similar webpage designs, operations processes, and patient characteristics. Therefore, it is unlikely that the unobservable physician-specific time-varying factors that matter for the performance of physicians are platform-specific. Taken together, these results suggest that unobservable physician-specific factors, if there are any, should not play a significant role in the relationship between physicians' team participation and their performance (measured by the number of satisfied reviews) in individual consultations.

Table B.2 Robustness Checks with a DD Model on Haodf

Dependent Variable:	(1) <i>SatiNum^H</i>
<i>After^H</i>	-0.003 (0.005)
<i>SatiCumNum^H</i>	0.000 (0.000)
<i>PatientNum^H</i>	-0.000 (0.001)
<i>AvgResponseLen^H</i>	4.88e-08 (5.25e-08)
<i>Tenure^H</i>	-0.000 (0.000)
Time FEs	YES
Physician FEs	YES
Adjusted R ²	0.103
Observations	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses. *SatiCumNum^H* is the cumulative number of satisfied reviews physician i received before day t on Haodf, *PatientNum^H* is the number of individual patients of physician i over the last 10 days before day t on Haodf, *AvgResponseLen^H* is the average length of physician responses of physician i over the last 10 days before day t on Haodf, and *Tenure^H* is the number of days since physician i registers on Haodf before day t .

Appendix C. Placebo Test

To further verify the robustness of our result and the parallel-trend assumption underlying the DDD approach, we conduct a placebo test. To do so, we first re-estimate the DDD model in Equation (2) with a shorter timeframe (two quarters before joining consultation teams to two quarters after joining, as illustrated in Figure C.1). Next, following previous studies (e.g., Jung et al. 2019, Kumar et al. 2019), we shift $After_{it}$ (the binary variable indicating whether treated physicians have joined consultation teams) to one quarter earlier. We use the consultation experience of the matched physicians before the treated physicians joined consultation teams and set the $After_{it}$ variable to 1 in the first quarter before their treated date. $After_{it}$ still equals 0 in the second, third, and fourth quarters before their treated date. Note that the shifted $After_{it}$ arbitrarily moves one quarter earlier before the treated physicians actually joined consultation teams. Therefore, if we rerun the DDD estimation using the newly defined variable $After_{it}$, its coefficient estimate should be insignificant DDD.

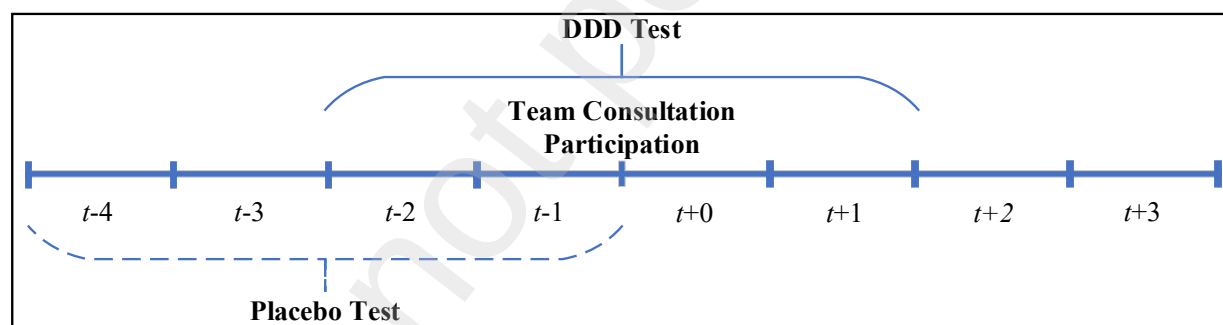


Figure C.1 Time Periods for the DDD and Placebo Tests

Table C.1 reports the results from the two estimations. For the DDD test with the unshifted $After_{it}$ indicator (in Column 1), the DDD estimate remains positive and significant. For the placebo test, the DDD estimate is insignificant (in Column 2), which is consistent with our expectation. Taken together, these results further support the parallel-trend assumption.

Table C.1 A Placebo Test

Specification	(1)	(2)
	DDD with a shorter time widow	Placebo test
Dependent Variable:	$\Delta SatiNum$	$\Delta SatiNum$
<i>After</i>	0.037*** (0.013)	-0.013 (0.010)
$\Delta SatiCumNum$	0.002* (0.001)	-0.002 (0.001)
$\Delta PatientNum$	0.006 (0.004)	0.004 (0.003)
$\Delta AvgResponseLen$	2.02e-08 (2.19e-07)	-6.34e-08 (9.42e-08)
<i>Tenure</i>	-0.000 (0.000)	0.000 (0.000)
Time FEs	YES	YES
Physician FEs	YES	YES
Adjusted R ²	0.088	0.080
Observations	85,441	64,763

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

Appendix D. Additional Analyses to Alleviate Self-Selection Concerns

D.1. A Hurdle Model

We conduct three sets of analysis to alleviate self-selection concerns. First, we account for confounding factors that may affect the likelihood of a physician being accepted into a team by other team members through a hurdle model. The hurdle model consists of a binary choice component that accounts for a physician's decision to join a team and a second component that captures the impact of team participation on individual consultations. For the binary choice component, we include physicians' offline and online characteristics. Specifically, we use the professional title and hospital level to measure their offline characteristics, as high titles and elite hospitals indicate physicians have a high level of professional capacity. We use physicians' tenure and the number of individual patients served over the last 10 days before day t on WeDoctor to capture their commitment or engagement on a platform. The rationale is that a high level of professional capacity and engagement can affect their chances of being accepted by other team members. The estimation results from the hurdle model are presented in Table D.1. We can observe that the estimate for the coefficient of $After_{it}$ is positive and significant, which is consistent with the finding from Table 3.

Table D.1 A Hurdle Model

Dependent Variable:	(1)
	$\Delta SatiNum$
Second Component	
<i>After</i>	0.035** (0.011)
$\Delta SatiCumNum$	0.001*** (0.001)
$\Delta PatientNum$	0.011*** (0.001)
$\Delta AvgResponseLen$	-3.47e-08 (6.94e-07)
<i>Tenure</i>	0.000*** (0.000)
Binary Choice Component	
<i>Professional Title</i>	-0.096*** (0.009)
<i>Elite Hospital</i>	0.215** (0.085)
<i>Tenure</i>	0.000*** (0.000)
$\Delta PatientNum^W$	0.073*** (0.002)
Time FEs	YES
Physician FEs	YES
Log-likelihood	-16,932.6
Observations	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

D.2. A Look-Ahead Analysis

We conduct a look-ahead analysis, where we use physicians treated earlier as the treatment group and those treated later as the control group. As members in both groups ultimately join a team, such an analysis further alleviates self-selection concerns (Huang et al. 2021). Among the 292 treated physicians, we use 146 physicians who joined teams earlier as the treatment group, and the other 146 physicians as controls. For the treatment group, we use their experience on the two platforms over two years, i.e., one year before (if any) and one year after joining teams (if any), as the study period. For the control group, we use their experience before joining consultation teams as the study period. The results are shown in Table D.2. The estimate for the coefficient of $After_{it}$ is positive and significant, which is consistent with that from Table 3.

Table D.2 Robustness Checks with A Look-Ahead Strategy

Dependent Variable:	(1)
	$\Delta SatiNum$
<i>After</i>	0.049** (0.018)
$\Delta SatiCumNum$	-0.001** (0.000)
$\Delta PatientNum$	0.013*** (0.003)
$\Delta AvgResponseLen$	2.95e-07 (2.21e-07)
<i>Tenure</i>	-0.000 (0.000)
Time FEs	YES
Physician FEs	YES
Adjusted R ²	0.165
Observations	116,276

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

D.3. An Instrumental-Variable Approach

We also employ an instrumental-variable approach to alleviate self-selection concerns. Specifically, we adopt the ratios of physicians who joined teams in the same hospital (perhaps in different departments) and the same department (perhaps in different hospitals) as the instrumental variables (denoted by $RatioHospital_{it}$ and $RatioDepartment_{it}$, respectively). The rationale is that physicians may learn about team participation from their peers in the same unit; their adoption decision may also be affected by their peers (Luo et al. 2018, Zhang et al. 2018). However, the prevalence of peers' adoption of team consultation should not directly influence the focal physicians' individual performance. We estimate the following 2SLS model:

$$After_{it} = \alpha_0 + \alpha_1 Instrument_{it} + \alpha_3 Z_{it} + \tau_{it}, \quad (D.1)$$

$$\Delta Y_{it} = \gamma_0 + \gamma_1 \overline{After}_{it} + \gamma_3 Z_{it} + \gamma_i + \epsilon_{it}, \quad (D.2)$$

where Equation (D.1) is the first-stage regression and Equation (D.2) is the second-stage regression. The independent variable, $Instrument_{it}$, corresponds to one of the two instrumental variables or both of them, depending on the specific analysis chosen. ΔY_{it} is the dependent variable, i.e., the difference in terms of the number of satisfied reviews between the two platforms for physician i on day t (i.e., $SatiNum_{it}^W - SatiNum_{it}^H$). Vector ΔZ_{it} is the set of control variables described in Equation (2).

In the first stage (i.e., Equation (D.1)), we find positive relationships between the two instrumental variables and the independent variable in Table D.3, confirming the relevance of the instrumental variables. We also verify the strength of the instrument variables using the Cragg-Donald Wald F statistic (5,624.130 for $RatioHospital_{it}$, 711.918 for $RatioDepartment_{it}$, and 2846.739 for them concurrently), which is much higher than the Stock-Yogo critical value for 2SLS relative biases of 5% (Stock and Yogo 2002). The Kleibergen-Paap rk Lagrange multiplier statistics also support the results (1,148.096, p -value < 0.001 for $RatioHospital_{it}$; 160.591, p -value < 0.001 for $RatioDepartment_{it}$; and 580.440, p -value < 0.001 for them concurrently) (Kleibergen and Paap 2006).

Table D.3 Instrument Relevance, First-Stage Regression of 2SLS

	(1)	(2)	(3)
<i>RatioHospital</i>	0.494*** (0.007)	-	0.480*** (0.007)
<i>RatioDepartment</i>	-	0.217*** (0.008)	0.068*** (0.008)
$\Delta SatiCumNum$	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
$\Delta PatientNum$	0.003*** (0.000)	0.002*** (0.000)	0.003*** (0.000)
$\Delta AvgResponseLen$	2.04e-06*** (1.33e-07)	2.61e-06*** (1.72e-07)	2.03e-06*** (1.33e-07)
<i>Tenure</i>	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
Time FEs	YES	YES	YES
Physician FEs	YES	YES	YES
Adjusted R ²	0.397	0.378	0.397
Observations	152,082	152,082	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

Table D.4 presents the results from the second-stage estimations with the estimate $\overline{After}_{it}$ to replace $After_{it}$ in the analysis. We find that the estimates for the coefficient of $\overline{After}_{it}$ are positive and significant in Columns 1-3, which are consistent with the observation from Table 3. Additionally, we conduct the Sargan-Hansen test to assess the overidentifying restrictions when using the two instrumental variables concurrently in Column 3. The χ^2 value is 1.921 (p -value = 0.166), confirming the validity of our instrumental variables in this analysis.

Table D.4 An Instrumental-variable Approach, Second-stage Regression of 2SLS

	(1)	(2)	(3)
Dependent Variable:	$\Delta SatiNum$	$\Delta SatiNum$	$\Delta SatiNum$
Instrumental Variable:	$RatioHospital$	$RatioDepartment$	$\frac{RatioHospital}{RatioDepartment}$
$\overline{After}$	0.136*** (0.014)	0.322*** (0.033)	0.079*** (0.028)
$\Delta SatiCumNum$	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
$\Delta PatientNum$	0.010*** (0.001)	0.010*** (0.001)	0.011*** (0.000)
$\Delta AvgResponseLen$	2.04e-06*** (1.33e-07)	-1.04e-06*** (1.36e-07)	-9.29e-08 (1.46e-07)
$Tenure$	0.000*** (0.000)	-0.000*** (0.000)	-7.34e-06 (2.85e-05)
Time FEs	YES	YES	YES
Physician FEs	YES	YES	YES
Observations	152,082	152,082	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

To address the forbidden regression issue pertinent to the instrumental-variable approach, we adopt the procedure recommended by Angrist and Pischke (2009). First, we use a Probit model to regress $After_{it}$ with the instrumental variables and the controls from the main analysis. Next, based on the results from the first stage, we calculate the predicted values of $After_{it}$, denoted as $\overline{After}_{it}$. Finally, we rerun Equation (D.2) with a regular 2SLS estimation approach with $\overline{After}_{it}$ replacing $After_{it}$. We present the results in Tables D.5 and D.6. We can see that the estimate for the coefficient of $After_{it}$ remains positive and significant, which is consistent with that from Table 3.

Table D.5 Instrument Relevance, First-Stage Regression Using Probit Estimation

	(1)	(2)	(3)
<i>RatioHospital</i>	2.188*** (0.477)	-	2.147*** (0.032)
<i>RatioDepartment</i>	-	0.986* (0.558)	0.165*** (0.039)
<i>ΔSatiCumNum</i>	0.003** (0.001)	0.004*** (0.001)	0.003*** (0.000)
<i>ΔPatientNum</i>	0.018 (0.021)	0.004 (0.020)	0.018*** (0.002)
<i>ΔAvgResponseLen</i>	8.93e-06 (6.04e-06)	1.20e-05** (6.04e-06)	8.94e-06*** (6.60e-07)
<i>Tenure</i>	0.001*** (0.000)	0.001*** (0.000)	0.001*** (1.67e-05)
Time FEs	YES	YES	YES
Physician FEs	YES	YES	YES
Pseudo R ²	0.251	0.221	0.251
Observations	152,082	152,082	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

Table D.6 An Instrumental-variable Approach, Second-stage Regression of 2SLS

	(1)	(2)	(2)
Dependent Variable:	<i>ΔSatiNum</i>	<i>ΔSatiNum</i>	<i>ΔSatiNum</i>
Instrumental Variable:	<i>RatioHospital</i>	<i>RatioDepartment</i>	<i>RatioHospital</i> <i>RatioDepartment</i>
<i>After</i>	0.064* (0.038)	0.085* (0.045)	0.063* (0.037)
<i>ΔSatiCumNum</i>	0.002 (0.001)	0.001 (0.001)	0.002 (0.001)
<i>ΔPatientNum</i>	0.010* (0.005)	0.011** (0.005)	0.010* (0.005)
<i>ΔAvgResponseLen</i>	-4.79e-07 (3.53e-07)	-8.52e-07 (5.33e-07)	-4.76e-07 (3.51e-07)
<i>Tenure</i>	4.50e-05 (8.14e-05)	2.84e-05 (8.34e-05)	4.58e-05 (8.12e-05)
Time FEs	YES	YES	YES
Physician FEs	YES	YES	YES
Adjusted R ²	0.092	0.092	0.092
Observations	152,082	152,082	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

Appendix E. Alternative Model Specifications

E.1. An Alternative Dependent Variable

In the main analysis, we use $\Delta SatiNum_{it}$, the difference in the number of satisfied reviews physician i received on day t , to measure physicians' performance in individual consultations. We also consider an alternative dependent variable $\Delta SatiRatio_{it(-10)}$, i.e., the difference in the ratio of satisfied patients to all individual patients over the last 10 days before day t for physician i between the two platforms. We operationalize this dependent variable using a longer timeframe for two reasons: (1) it takes time for patients to express satisfaction from receiving physician responses in consultations, and (2) physician may not receive patients or provide consultations on both WeDoctor and Haodf every day. Using $\Delta SatiRatio_{it}$ as the dependent variable in Equation (2), we replicate the DDD estimation. Table E.1 presents the results, which shows that the estimate for the coefficient of $After_{it}$ is positive and significant.

E.2. Additional Control Variables

In the main analysis, we use physicians' consultation experience over the last 10 days before day t to account for the differences between the two platforms. As a robustness check, we use more controls related to physicians' experience on the two platforms, which include the differences in the cumulative number of individual patients (i.e., $\Delta PatientCumNum_{it}$), the differences in the cumulative estimated earnings (i.e., $\Delta CumIncome_{it}$, calculated based on the consultation prices and number of consultations conducted), and the differences in the cumulative questions answered (i.e., $\Delta CumQuestionNum_{it}$) between the two platforms. Table E.2 presents the results, which are consistent with those shown in Table 3.

Table E.1 Robustness Checks with Different Variable Constructions

Dependent Variable:	(1) <i>ΔSatiRatio</i>
<i>After</i>	0.004* (0.002)
<i>ΔSatiCumNum</i>	0.000 (0.000)
<i>ΔPatientNum</i>	-0.001* (0.001)
<i>ΔAvgResponseLen</i>	-5.06e-08 (8.60e-08)
<i>Tenure</i>	6.23e-06 (2.53e-05)
Time FEs	YES
Physician FEs	YES
Adjusted R ²	0.024
Observations	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

Table E.2 Robustness Checks with More Controls

Dependent Variable:	(1) <i>ΔSatiNum</i>
<i>After</i>	0.029** (0.014)
<i>ΔSatiCumNum</i>	0.002* (0.001)
<i>ΔPatientNum</i>	0.012** (0.005)
<i>ΔAvgResponseLen</i>	-1.39e-08 (2.06e-07)
<i>Tenure</i>	0.000 (0.000)
<i>ΔPatientCumNum</i>	-0.004** (0.002)
<i>ΔCumIncome</i>	0.000** (0.000)
<i>ΔCumQuestionNum</i>	0.000 (0.000)
Time FEs	YES
Physician FEs	YES
Adjusted R ²	0.101
Observations	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

E.3. Alternative Time Windows

We use a two-year window, i.e., one year before and one year after joining consultation teams, in the main analysis. There may be potential contemporaneous shocks in the one-year treatment time window. To mitigate concerns about the potential impact of other shocks, we further test the robustness of our results using different time windows, and present the results in Table E.3. We can observe that the estimates for the coefficient of $After_{it}$ are all positive and significant.

Table E.3 Robustness Checks with Different Time Windows

	(1)	(2)	(3)	(4)	(5)
Time Window:	30 days	60 days	90 days	120 days	150 days
Dependent Variable:	$\Delta SatiNum$	$\Delta SatiNum$	$\Delta SatiNum$	$\Delta SatiNum$	$\Delta SatiNum$
$After$	0.039*** (0.013)	0.031** (0.013)	0.026** (0.011)	0.030*** (0.010)	0.033*** (0.012)
$\Delta SatiCumNum$	-0.002 (0.006)	0.005 (0.004)	0.007* (0.004)	0.004* (0.002)	0.002* (0.001)
$\Delta PatientNum$	0.014*** (0.003)	0.005 (0.004)	0.004 (0.004)	0.005 (0.004)	0.006 (0.004)
$\Delta AvgResponseLen$	-3.31e-08 (1.90e-07)	-2.72e-07 (2.46e-07)	-1.38e-07 (1.47e-07)	-1.07e-07 (1.35e-07)	2.52e-08 (2.07e-07)
$Tenure$	-0.000 (0.000)	-0.000 (0.000)	-0.000** (0.000)	-0.000 (0.000)	-0.000 (0.000)
Time FEs	YES	YES	YES	YES	YES
Physician FEs	YES	YES	YES	YES	YES
Adjusted R ²	0.076	0.074	0.117	0.105	0.095
Observations	15,492	30,077	44,266	58,122	71,991

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

E.4. Concerns on Sample Size

Since our main analysis is based on a sample of 338 matched physicians, we conduct several analyses to show that the limited sample size does not constitute a serious issue for our DDD estimation. First, we randomly select 150 physicians (i.e., 75 from the treatment group and 75 from the control group) as a subsample to estimate Equation (2). The estimate for the coefficient of $After_{it}$ in Column 1 of Table E.4 is positive and significant. Second, we use bootstrapping to compute the standard errors (10,000 times, with replacement) in the DDD estimation (Pu et al. 2022). From Column 2 of Table E.4, we can see that the coefficient of $After_{it}$ is positive and significant, and the results are consistent with those in Table 3.

Table E.4 Robustness Checks with Power Analysis

Specification Dependent Variable	(1)	(2)
	A subsample $\Delta SatiNum$	Bootstrapping $\Delta SatiNum$
<i>After</i>	0.057** (0.028)	0.042*** (0.004)
$\Delta SatiCumNum$	0.002 (0.001)	0.002 (0.000)
$\Delta PatientNum$	0.013 (0.009)	0.011** (0.001)
$\Delta AvgResponseLen$	1.58e-07 (8.71e-07)	-2.52e-08 (1.15e-07)
<i>Tenure</i>	0.000** (0.000)	8.20e-05 (0.000)
Time FEs	YES	YES
Physician FEs	YES	YES
Adjusted R ²	0.107	0.092
Observations	66,931	152,082

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses.

Appendix F. Alternative Matching Methods

In the main analysis, we use CEM to ensure the same number of physicians in the treatment and control groups. We also consider alternative matching methods to check the robustness of our results. First, we conduct another CEM without restricting the number of physicians being the same in the treatment and control groups, and 252 treated physicians and 183 controlled physicians are matched. With these 435 physicians, we re-estimate our DDD model in Equation (2) with the same model specifications, research period, and controls, and the results are presented in Column 1 of Table F.1.

Second, in addition to the matching variables explained in Appendix A, we include the following physician characteristics to control for the differences between the two platforms in CEM while restricting the number of physicians being the same in the treatment and control groups: the rank of the city (based on its business attractiveness) where the physician's hospital is located, the physician's gender, and the medical department that the physician is affiliated with. 268 physicians are matched, and half of them are from the treatment group. The results from these physicians are shown in Column 2 of Table F.1.

Third, we use one-to-one propensity score matching (PSM) without replacement, in which the propensity score of a physician is predicted and then one treated physician is matched with one controlled physician with a similar propensity score (Caliendo and Kopeinig 2008). Using the same matching variables as in the main analysis, there are 181 treated physicians, and 181 controlled physicians matched for the DDD estimation. The results are presented in Column 3 of Table F.1.

Fourth, we use PSM with replacement to allow one controlled physician to be matched to more than one treated physician. Under this matching scheme, 292 records of controlled physicians are matched to 292 records of treated physicians, and the DDD estimation results based on these 584 physicians are shown in Column 4 of Table F.1.

To facilitate comparison, we also include physicians in the full sample to replicate the DDD estimation and present the results in Column 5 of Table F.1. Overall, we find that the estimate for the coefficient of $After_{it}$ is positive and significant, and the results are consistent with those in Table 3.

Table F.1 Tests of Alternative Matching Approaches

Matching Approach	CEM (without <i>k2k</i>)	CEM (with <i>k2k</i> and more matching variables)	One-to-One PSM (without replacement)	One-to-One PSM (with replacement)	No Matching
	(1)	(2)	(3)	(4)	(5)
Dependent Variable	$\Delta SatiNum$	$\Delta SatiNum$	$\Delta SatiNum$	$\Delta SatiNum$	$\Delta SatiNum$
<i>After</i>	0.030*** (0.010)	0.022** (0.010)	0.047** (0.019)	0.045*** (0.013)	0.034*** (0.012)
$\Delta SatiCumNum$	0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.000 (0.001)	1.52e-05 (0.000)
$\Delta PatientNum$	0.019** (0.005)	0.005 (0.000)	-0.004 (0.007)	0.0192*** (0.004)	0.011*** (0.002)
$\Delta AvgResponseLen$	1.29e-07 (2.95e-07)	6.06e-06 (6.35e-06)	2.54e-07 (2.01e-07)	2.55e-07 (2.20e-07)	6.34e-08 (5.17e-08)
<i>Tenure</i>	0.000 (0.000)	-0.000** (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (4.08e-05)
Time FEs	YES	YES	YES	YES	YES
Physician FEs	YES	YES	YES	YES	YES
Adjusted R ²	0.102	0.007	0.165	0.154	0.078
Observations	151,767	80,965	101,339	176,991	2,556,204

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Robust standard errors are in parentheses

Appendix G. Tests of Alternative Explanations

We conduct additional analyses to rule out other alternative explanations. Table G.1 summarizes the results. The first alternative explanation concerns the distinct roles (leaders vs. non-leaders) within teams. To investigate the plausibility of such an explanation, we replicate the DDD estimation on two subsamples that consist of team leaders and non-leaders, respectively. We can observe from Columns 1 and 2 that the coefficient estimates are positive and significant for both subsamples, suggesting that the effect of team participation persists regardless of leadership status. We further create a dummy variable $Leader_i$ to indicate the leadership role of physician i within a team, and include it as a moderator in the DDD model to rerun the estimation. Column 3 shows the coefficient of the interaction term is insignificant, hence ruling out the possibility that the observed spillover effect is primarily driven by the status difference in a virtual consultation team.

Another alternative explanation we consider is that physicians that joined a team may pay more attention to the treatment platform, making them more responsive, which leads to a shorter response time to patients' inquiries. To test the plausibility of this explanation, we use the difference in physicians' response time in individual consultations across the two platforms, i.e., $\Delta Responsetime_{it}$, as the dependent variable in the DDD model described in Equation (2). It follows from Column 4 that the coefficient is not significant, suggesting the attention-based explanation is unlikely to serve as the major driver for the observed effects.

Table G.1 Tests of Alternative Explanations

Sample	Leaders	Non-leaders	Full	Full
	(1)	(2)	(3)	(4)
Dependent Variable	$\Delta SatiNum$	$\Delta SatiNum$	$\Delta SatiNum$	$\Delta Responsetime$
<i>After</i>	0.041** (0.018)	0.043** (0.017)	0.030 (0.020)	0.004 (0.004)
<i>After</i> × <i>Leader</i>	-	-	0.042 (0.046)	-
$\Delta SatiCumNum$	0.002 (0.002)	0.002** (0.009)	0.002 (0.001)	3.52e-05 (6.02e-05)
$\Delta PatientNum$	0.013 (0.011)	0.011* (0.006)	0.011** (0.005)	0.003** (0.001)
$\Delta AvgResponseLen$	-1.20e-05 (2.62e-05)	9.79e-09 (2.13e-07)	-9.76e-09 (2.06e-07)	5.58e-07** (2.73e-07)
<i>Tenure</i>	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)
Time FEs	YES	YES	YES	YES
Physician FEs	YES	YES	YES	YES
Adjusted R ²	0.088	0.097	0.092	0.008
Observations	47,093	104,989	152,082	141,781

Notes: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$; Robust standard errors are in parentheses. There are fewer observations in Column (4). This is because some physicians did not respond to any individual patients on some days, in which case the difference in the response time cannot be properly calculated.

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